

Research Topic "The influence of soil moisture on forest windthrow-A pre study on data availability and suitability"

Prakash Gautam

Faculty of Forest and Environment
Forest Information Technology,
HNEE Eberswalde

Supervisor:
Nicole Albert
Dr. Evelyn Wallor

Abstract

Soil moisture plays a significant role in the environment. Several factors like water, carbon dioxide, mist, and energy are connected to the earth's soil moisture and ecology. This study intends to notice the soil moisture data from the remote sensing perspective from satellites, mainly Sentinel -1 C and Soil Moisture Active Passive (SMAP), non-identical satellites having discrete spectral and spatial resolutions where data are tested on three distinct locations of two windthrow storm events happened on the year 2017 and 2018 on the European continent mainly in the German Federal States. The data footprint approach was to visualize the effectiveness of spatial and temporal efficacy to the user for a specific time frame. The method was entirely data-driven from satellite and ancillary data from precipitation events. The result shows differences in soil moisture of satellite data concerning its availability and acquisition.

Introduction

Fundamentally soil moisture is a major state variable in the environment regulating the flow of water, energy, evaporation, infiltration, and carbon flux between the land surface and the atmosphere. The soil moisture is the water content in the vadose zone, in which hydrological, physical, chemical and biological processes related to the atmosphere and the land surface process occur (Meyer et al., 2021). The knowledge of soil moisture helps to evaluate humidity and temperature in the global water and energy cycle (Ahlmer et al., 2018). Considering soil moisture's importance, data have broad applicability in multiple backgrounds from the precedent numerous satellite observations in the universal domain. Those domains are climate variability, land ambience, multiple chemical cycles and land surface modelling, aridity and weather casting (Cai, 2016). A review and evaluation of remote sensing soil moisture data are vital in the area, especially soil cover and topography, where factors like soil that cover vegetation and land affect the soil-moisture estimation. The decreasing soil moisture indicates drought, where water availability is altered to vegetation and soil, negatively affecting soil properties and influencing other numerous soil characteristics. The drained soil moisture affects evaporation, cooling, and air temperature increment showing anomalies in heatwaves amplifying and obliging on atmosphere and boundary layers (Miralles et al., 2014 a). In the earlier studies carried out for soil moisture study, satellite products having huge swath capturing temporal scale were observed. However, the products were only available at the continental level lacking fine spatial details and not supporting details on the local level. The most high-resolution satellite generating footprint in particular location sporadically, with inadequate visit creating space gap and

failing to capture short-term variations like soil moisture and precipitation and considering the underlying factor necessity of short-repeating coarse scale data product needed with temporal stability to get the sub-pixel information on the local level (Bauer-Marschallinger et al., 2018). Additionally, remote sensing data are not always accurate and precise. There is a frequent issue for satellite data in spatial and vertical scale discrepancies. During sensing, there are ecological and temporal abrupt changes on the ground and the soil profile in very minute scale. The local attribute factor like geolocation, distance, neighboring environment, interaction with topographical features, provides detailed accuracy in the data and interacting agent (Arlinghaus, 2019). Therefore, right analysis of the correct soil moisture information is needed. Using satellite imagery can help in decision-making, mitigate extreme events, and significantly increase efficacy in managerial and measurements. In the scientific study, it is necessary to investigate and carry out data analysis to embark on the final research project. Since data must be generated, tested, and validated during scientific research.

Research Question

This research aims to gather knowledge about the soil moisture value on the windthrow damage polygons in the study region. Specifically, it strikes to investigate humidity on the study location with the help of globally recognized data sets from Land Copernicus and NASA during the storm phase as well as the pre-post precipitation phase in the affected area. It tries to investigate spatial relationship between the windthrow polygons depending upon the soil moisture datasets. Furthermore, it tries to analyze difference soil moisture grid pixel values and observe footprints during the event. Additionally, it aims to evaluate, compare, and match against satellite data footprint for the survey in the timeframe to achieve the best possible visualization. It aims to collect information on total precipitation of local weather stations to validate remote sensing data during the event. Furthermore, research regarding the upscaled, downscale, or both data performing the best suitability of remote sensed data with various spatial and temporal resolution generation of footprint analysis for time events

Materials and Methods

Study area and storm event

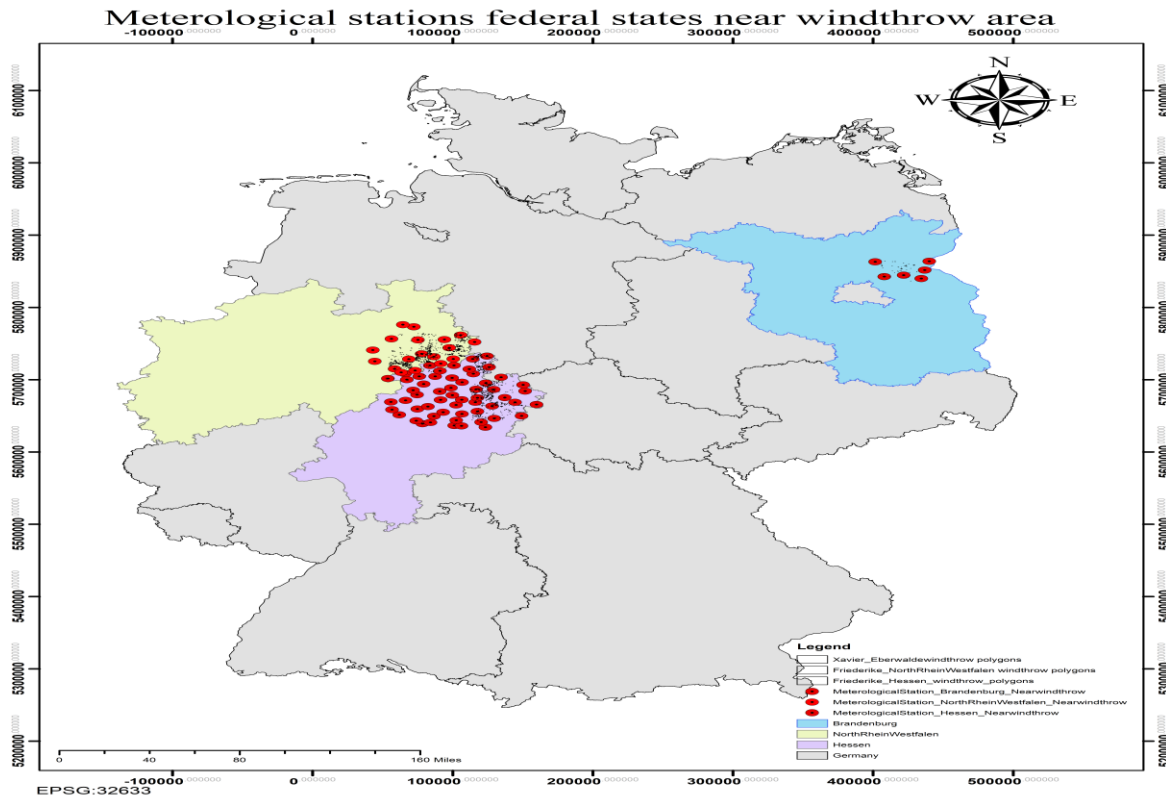


Figure 1: Overview of the study location of three federal states, including meteorological weather stations and windthrow damage polygons.

As observed in the map, the study location consists of three different places in the federal states of Germany. Brandenburg (BB), where a windstorm took place in the year 2017 on the first week of October near the Eberswalde region, a strong windthrow "Xavier" effect was observed on the forest region. The windthrow Xavier passed over the northern part of Germany at 120 km per hour, demolishing the big city. Transport and infrastructure with significant economic damages also caused substantial damage to various forest regions, especially in Berlin, and Brandenburg, where almost 40,000 trees were partially or damaged with numerous degrees into complete failure (Gross, 2018). Furthermore, the wind had shown higher turmoil on the region. The Brandenburg forest administration calculated that 1 million of subsidiary timber has to be cleared, translating to 1.5 to 2 million individual trees being tossed in Brandenburg (Gross, 2018). On January 2018, windthrow another storm event named Federike hit again in the lower western part of Germany, primarily affecting North Rhein Westfalen (NRW) and Hessen. The wind had a speed of 203 kilometres per hour, destroying trees, roads, and other infrastructure. Those extreme wind events create and develop wind power in the forest area, destroying trees and creating several windstorms damaging areas with several hectares in the federal states. At the time of the event, trees were swinging back and forth until they broke

under wind pressure or were tossed down along with their roots rather than falling suddenly (Vautard et al., 2019).

Workflow Diagram:

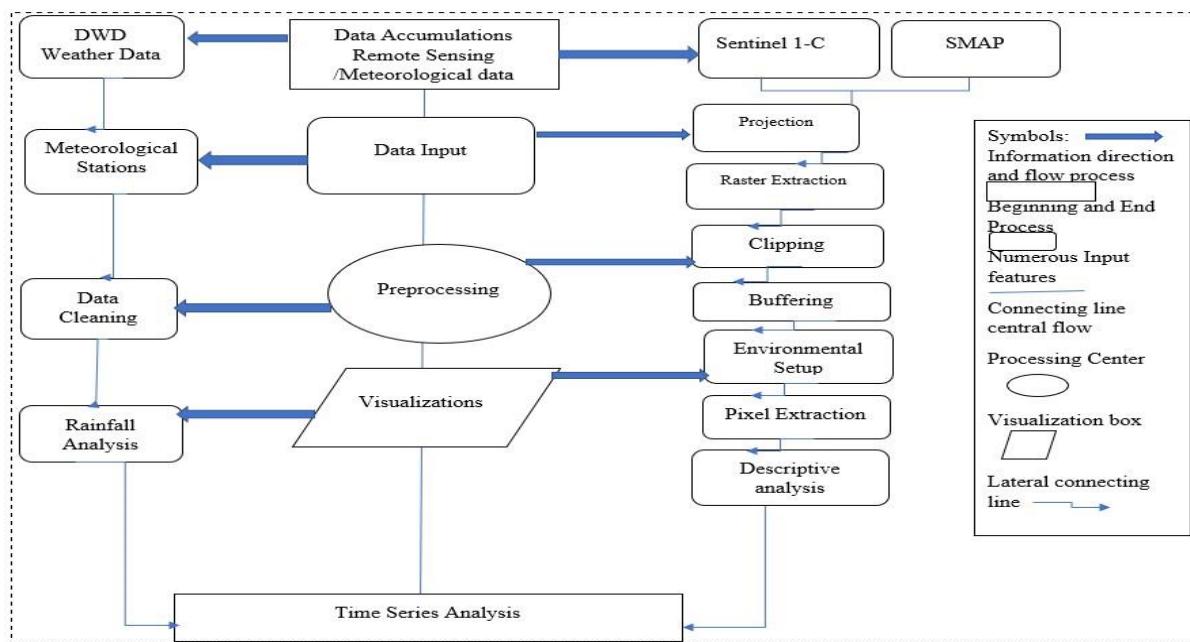


Figure 2: Workflow diagram of the research project -The middle part consists of the main flow where the right part remote sensing data flow takes place, whereas the left side is ancillary data.

Overview of Germany Forest

The forests in Germany are located near agricultural land, water streams, low mountains, and alpine terrain. Despite of 32% extension of woods on this land shows much forest cover in Germany however it does not always support healthy forests in equal proportion. Various sources show that predominated tree species showing much diversity of coniferous species dominating forested areas. In contrast, later followers were broad leaves and mixed forest continuing at 31% and 13 %. The other species are spruce 25%, pine 3%, broad leaves beech in 16%, and Oak in 11% (Holzwarth et al., 2020).The most common tree species found in Germany are as Common Spruce, Common Pine, Copper Beech, Sessile, Oak, English Oak, Common Birch, Common Ash, Black Alder, European Larch, Douglas Fir, and Sycamore Maple shows the distribution frequency of tree species among the various federal states in Germany.

In-situ Datasets

The In-situ datasets are reference data containing information on environmental and geographical accuracy. Ancillary datasets are practiced in remote sensing studies as a secondary source for the testimonial data to schedule historical rainfall, water abstraction, and meteorological studies (Hochstöger, 2016). The datasets also provide additional information on periods as well as season. In the case of soil moisture, some relevant information like soil description, soil depth, infiltration rate, season, crops, and geographical area information is needed. For in-situ datasets, field data sets called 'Ground Truth Data' collected directly from field observation to determine the real and exact attributes of study location to compare and interpret with remote sensing datasets that can be mainly helpful for establishing point and matching positions of satellite imagery. Usually, weather and climate monitoring stations collect data on the ground for more accessibility and compatibility as secondary data.

Precipitation Datasets for Project

To address the research goal, precipitation data were assembled from the weather portal. The Deutscher WetterDienst (DWD), German National Meteorological Service storing various types of weather data in the weather portal. Since meteorological datasets are long-term observation series of precipitation events prepared for the suitability of the study area (Kreklow, Tetzlaff, Burkhard, & Kuhnt, 2020). Those data sets are obtained from the numerous type in various places of local meteorological stations. In addition, (DWD) and Climate Data Center (CDC) regulate collections, providing historical daily precipitation data from locally running weather stations from different federal states. The data set contains daily precipitation records and other information like station identification and its name. Additionally, rainfall data provides metadata where micro details like data acquisitions time, day, month, and year. Similarly, Other categories listed were station's height, relocation, periodic adjustment, an algorithm used during the measurement process, and establishment time of the stations.

Remotely Sensed Soil Moisture Datasets

The remotely sensed satellite-derived data depends upon the spatial and temporal resolutions. The product's usability must be implemented on the global, regional, and local analysis depending upon footprint availability for user necessity and requirements. Since microwaves wavelength ranges from 1 meter(m) to 1 millimeter (mm) having potential to perforate cloud cover, moisture, and dizziness, allowing waves to reach on the ground with the backscattering return. However, mediocre spatial resolution and lower temporal resolution obstruct the soil moisture measurements.

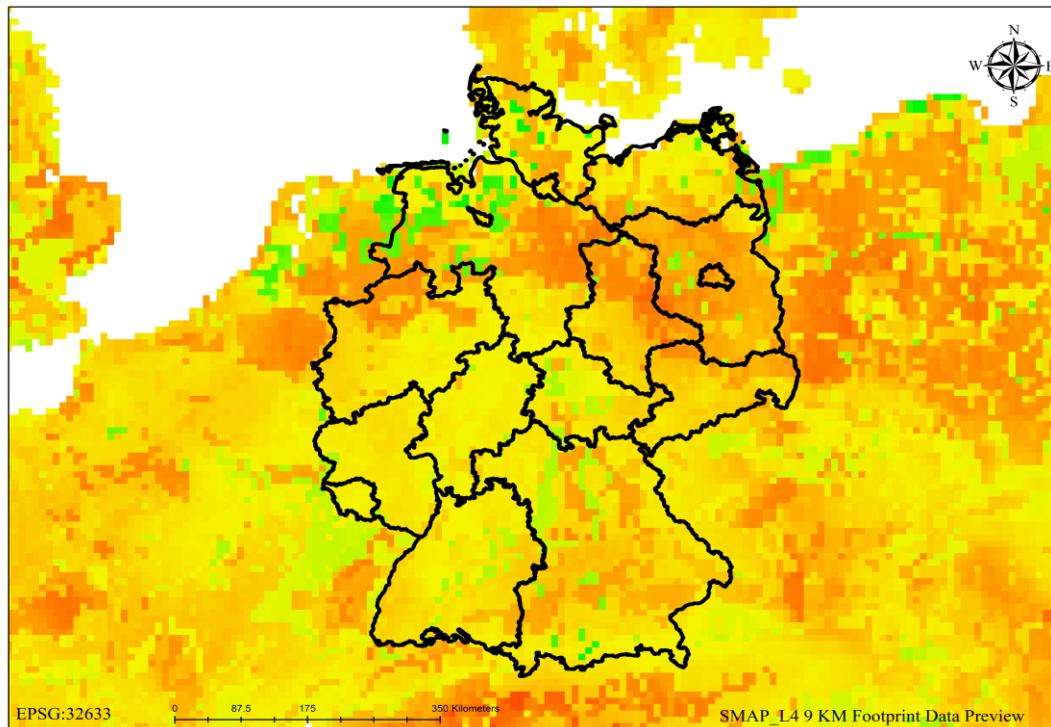


Figure 3: SMAP soil moisture footprint

Soil Moisture Active Passive (SMAP) Data

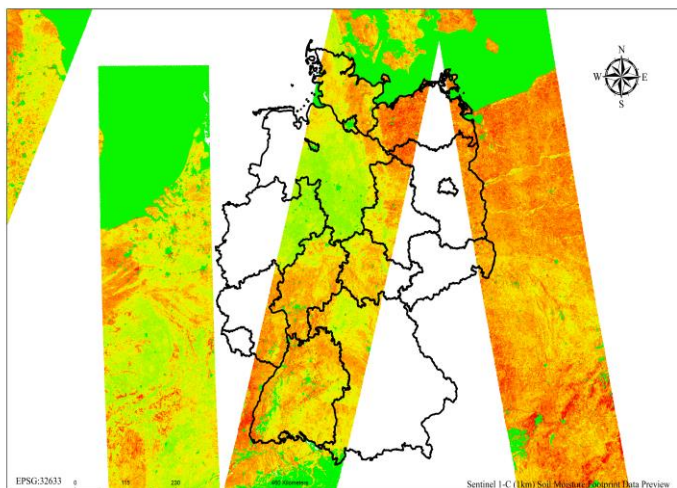
The SMAP is near a polar satellite with a sun-synchronized orbit having L-band radiometer (1.41GHZ), developed and launched by NASA in January 2015. The soil moisture active passive is used to monitor the soil moisture from its atmospheric orbit. It was established to understand the possible link between terrestrial water and energy concerning carbon cycles and global water energy fluxes reference land surfaces, predicting flood predictions and drought monitoring capacities. It also monitors related parameters of moisture, energy, and carbon associated with land topography as well as the carbon content of nature. The satellite data SMAP is a downscaled dataset from 36 km to 9 km resolution (Das et al., 2011, Das et al. 2014). The data of SMAP are designed in Hierarchical Data Format (HDF5) in formatted pattern with an accusation angle of 40 degrees and observed that the passive radiometer soil moisture SMAP where L band uses soil moisture radiometer setup for measuring 5 cm soil in numerous scales, obtaining from a passive L band radiometer and an active L band SAR (Entekhabi et al; 2014). The SMAP Consist of 4 different levels of data category, which fine-line 1 requirement of L4 SM product(Colliander, Jackson, & Bindlish, 2017). The level 4 SMAP soil data consists of surface soil moisture (5 cm) on the upper layer and rootzone soil moisture (0-100 cm) on lower surface. The same datasets contain additional products like soil temperature, evaporation fraction, radiation, and error estimation where soil moisture data values are provided in volumetric units (Reichle et al., 2015). The main products of SMAP are geophysical data, analysis data, and land model data. The analysis data is further categorized into forecast data and observation

data where soil moisture and temperature value are found. The spatial resolution of the radiometer is approximately 40 km, where the land surface model data are gridded into 9 km (Brodzik et al., 2012) with X and Y cells of 9 * 9 square kilometers in latitude and longitude. High-resolution SAMP provides a measurement reading of soil moisture in the top outer layer of the soil and inner layer. The most targeted application preferred by SMAP is rootzone moisture, which is not measured by SMAP, a model-based value is added on the level 4 data product to address the shortcomings of surface observations.

SMAP Rootzone Soil Moisture (RZSM) data

The (RZSM) data are derived by using an exponential filter model. The soil moisture of the root area has the primary role in balancing water and energy during transpiration and hydraulic redistribution. For hydrological variables like soil moisture, the L band is used in active and passive remote sensing because of its strong connection with the upper layer of soil. RZSM used both active and passive sensors, where the passive sensor provides a spatial resolution of (40-60) km in the agronomics activity. For monitoring purposes, only observing SSM alone is not enough because it can show disconnection between an upper-top surface and sub-surfaces, creating uncertainties in Rootzone Model (Das et al., 2019). For the best use data blending procedure is used for providing various uncertainties. The Rootzone model uses a large number of datasets so Kalman Filter Model is best practiced for the SM profile estimation (Reichle, Lannoy, Koster, Crow, & Kimball).

Sentinel 1-C Surface Soil Moisture Datasets



The Sentinel 1-C soil moisture data is a combined product of both Sentinel S-1 A and S-1 B data. Generally, a single Sentinel satellite has revisit time of once in 12 days, whereas consecutive data accessibility can be achieved in six days (Balenzano et al., 2021; Benninga, van der Velde, & Su, 2020). When both data are taken in a single frame, the revisit time reduces, resulting in data obtained

with 1.5 to 4 days over Europe whereas 3-6 days globally, depending on the location and footprint availability. The data used in the project are Land Copernicus Sentinel datasets remodeled by ESA from its hub (Peng, Loew, Merlin, & Verhoest, 2017). The soil moisture raster data of ESA are designed (Network Common Data Form) in N-CDF file format of Sentinel 1-C, pre-processed grid raster files being modelled with soil moisture values from ESA (Richard Kidd, 2021) and having a spatial resolution of 0.0012 degrees and operated from Copernicus Global Land Operation (CGLS).

The datasets are daily composite images (orbit) assembled with masks and noises (Bauer-Marschallinger, 2019). SSM 1 km data values, consist various quality information in different categories like dry and wet references denoted by codes like 241-242, 251-253 (extremely minimum and maximum) and different masking procedure for performance, improvement of the data quality and better accuracy. The masking is done in higher water sources area, slope region, and sensitive zone like human residential region. The purpose was for better modelling process and higher performance for the improvement of the data quality and its accuracy. The N-CDF file formats are file containing multiple variables like temperature, humidity, pressure, wind speed, and direction, basically storing format for meteorological and climatic metadata conventions (Kimerling, 2002).

Methodology (Data Preprocessing)

The methodological workflow of this research is based on the principle of correct data assemblage i.e., remote sensing data and precipitation data to start the model. The obtained data sets are introduced into flow series with the correct input features-processing, visualization, and outcome as time series analysis. The satellite (SMAP) level 4 data and Sentinel 1-C data were obtained from the portal for six days daily depending upon the first availability. Based on data unit, the satellite datasets were projected and reprojected universal transversal meta core (UTM) in EPSG 32633, clipped, reshaped, and stored in tiff format. Additionally, raster soil moisture values were brought into same unit. For storm damaged windthrow polygons of the three federal states having three different Coordinate Reference System (CRS) were reprojected on the same CRS EPSG 32633. Whereas scattered polygons were analysed by creating spatial boundary buffers from the centroid or mean distribution. The unalienated geometry were fixed for testing and windthrow polygons were tested twice by buffering to understand the geophysical pattern for spatial pattern analysis i.e. overlap or intersect, as well as to extract soil moisture value grid pixel value from satellite data. The polygon count and intersection were also conducted to know the of interaction cluster count so that individual windthrow polygons characteristics can be studied depending upon their geopositioned and damage area.

Buffering

Several types of buffering were carried out during the analysis, i.e., zero buffering for the polygons having problems with unalienated geometry and multiple ring buffer to understand the spatial relation of windthrow patches. Whereas point buffering was performed for the pixel extraction process of every single polygon. Medium buffering for the collection of higher topological features and understanding clustering and intersection in between the polygons with

possible other new features. Additionally, larger buffers for extracting soil moisture pixel values from datasets containing soil moisture grid. The outer layer buffering was created depending on the distribution of windthrow polygons in three federal states near study area. The soil moisture values were extracted from the raster grid from two different satellites with three soil moisture data sets, i.e. Surface Soil Moisture 5 cm from Sentinel 1-C, 5 cm outer surface layer SAMP and 100 cm Rootzone layer. For each storm damage polygon average pixel value is extracted from raster grids. The descriptive analysis was performed for both soil moisture and rainfall data and comparison performed for soil moisture data footprint from station lying on nearby storm damage areas and short time series was performed and visualized.

Results and Visualization

Soil moisture value in polygons

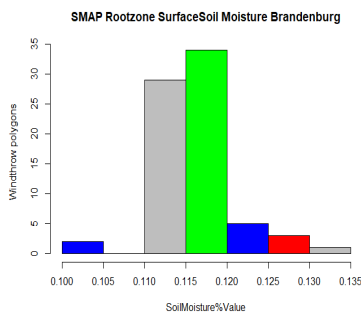


Figure 1 Rootzone Brandenburg

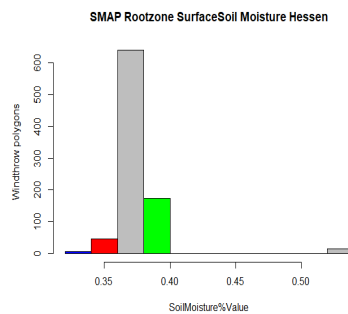


Figure 2 Rootzone Hessen

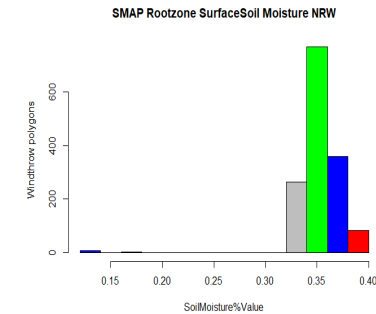


Figure 3 SMAP Rootzone NRW

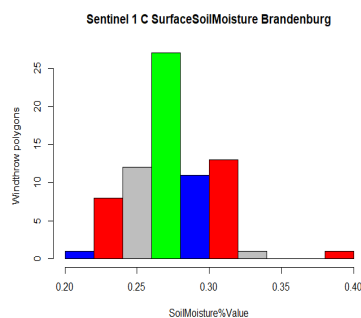


Figure 4 Sentinel-1 C Brandenburg

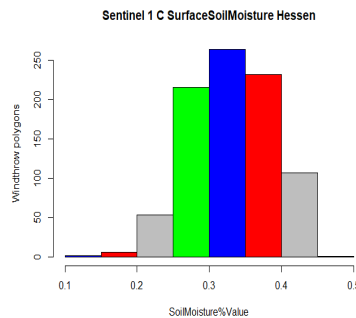


Figure 5 Sentinel 1 C SSM Hessen

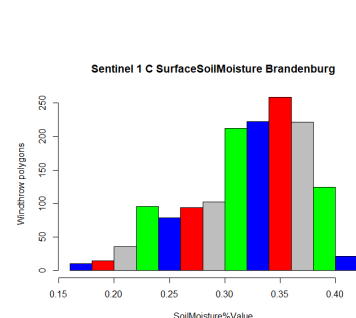


Figure 6 Sentinel-1 C SSM NRW

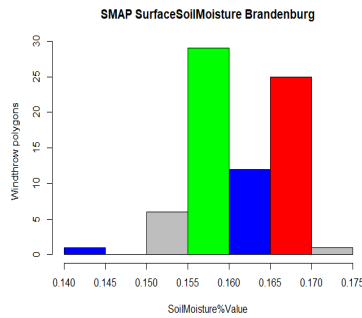


Figure 7 SMAP SSM Brandenburg

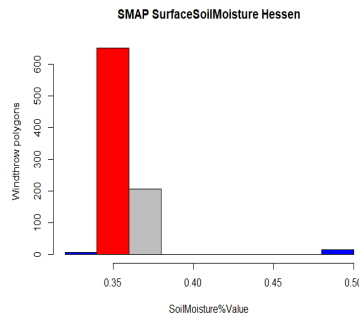


Figure 8 SMAP SSM Hessen

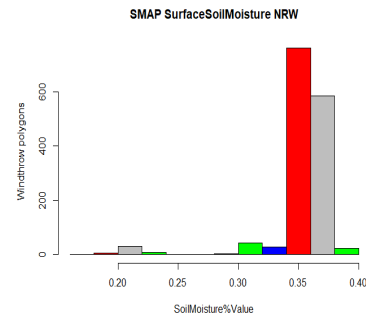


Figure 9 SMAP SSM NRW

On the above figure 1-3 represent rootzone values of individual polygons, 4-6 represent Sentinel 1 - C soil moisture values and 7-9 represents SMAP surface soil moisture. For all those patches of 6 days mean value of SMAP and first availability footprint mean value Sentinel 1 -C value was noted. During analysis it was noticed that Sentinel 1- C the soil moisture values were much more diverse and classified with spatial pattern whereas SMAP surface soil moisture and rootzone were less distributed.

Comparison soil moisture visualization on soil map series on Federal States during wind event.

The results of soil moisture data were compared in different time frames, Xavier storm from the first day till 6th October 2017, whereas Frederika 11th to 16th January for the SMAP data. The soil map generates temporal mean pixel values with different satellite sensors with different resolutions. The pixel values are set from lower to higher, where 0.01 is the lowest and 0.65 maximum in the soil moisture value. In soil map, inside and outer zone are demarcated with different buffer that shows the soil moisture of the storm area.

In figure 3, soil map shows rootzone soil moisture observation related to windthrow areas with different time frames. Inside and outside of buffer zone, the grid pixel shows spectral difference on 3 storm locations where during Xavier storm event, showing RZSM values during on the storm phase stating pixel value shifting from the raster shows changes on wetness on the pixel grid.

In figures 4 The soil map shows surface soil moisture of 5 cm on the outer layer of SAMP, pixel shows a higher soil moisture value. It can be assumed that due to infiltration, the soil moisture value is higher in the Frederika storm event. The increasing number of moistures toward optimum when the heavier amount of precipitation.

In figure 5 displays the 1 km grid shows much finer details soil moisture pixel values on the windthrow location. The soil moisture value is maximum on the windthrow day and higher amount of precipitation.

SMAP rootzone soil moisture pixel value (0-100) cm

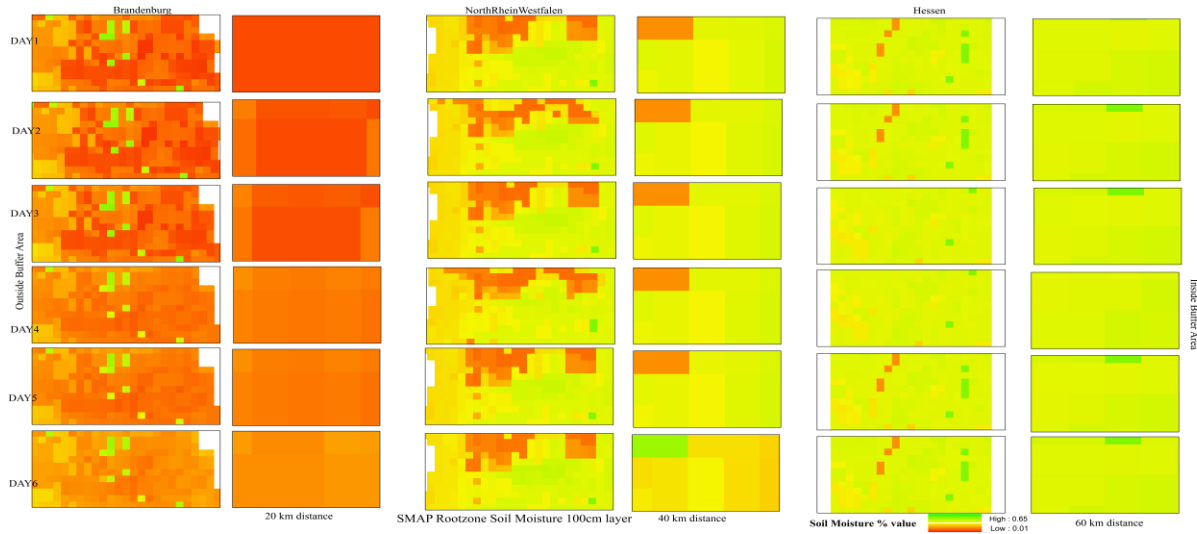


Figure 3: The soil moisture from SMAP satellite on 100 cm on upper layer during the storm event.

SMAP Rootzone soil moisture map: During analysis on Brandenburg, it was found that on day 1 to day 3 soil moisture found to be similar from day 4-pixel value changes to day 6 from darker to brighter. In NRW pixel values are similar in till day for where pixel of soil moisture on day 6 shows pixel values changes. In Hessen outer buffer zone was spotted higher on day 3 -4 having greater record on number soil moisture pixel count however rootzone moisture content does seem much more affected on the storm event of 5th and 6th day.

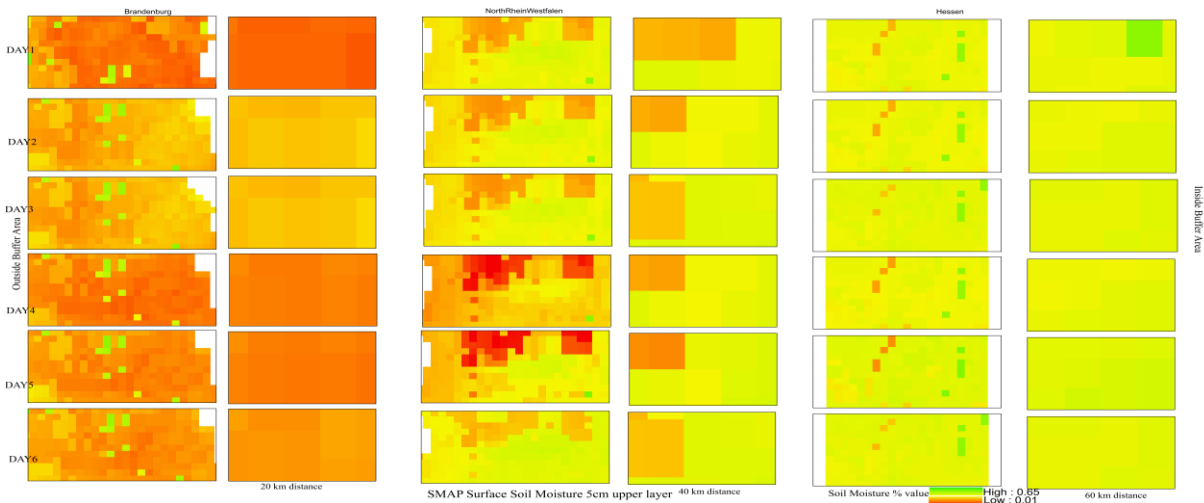


Figure 4: The soil moisture from SMAP satellite on 5 cm on upper layer during the storm event.

SMAP surface soil moisture: During analysis, In Brandenburg region shows day 2 and day 3 soil moisture pixel seems higher than first day, pixel count dropped on the 4th - 5th day found to be hollower to darker and increase their count on storm event. In NRW pixel values are recorded

higher on second and 4th which allows followed on 6th day whereas 4th and 5th day soil surface moisture was less on outer buffer area however increased their pixel on day 6th shows changes on the wetness .In Hessen outer buffer zone was spotted higher on day 3 -4 having greater record on number soil moisture pixel count however rootzone moisture content does seem much more affected on the storm event of 5th and 6th day

Sentinel 1 -C surface soil moisture grid pixel value 5(cm) soil map:

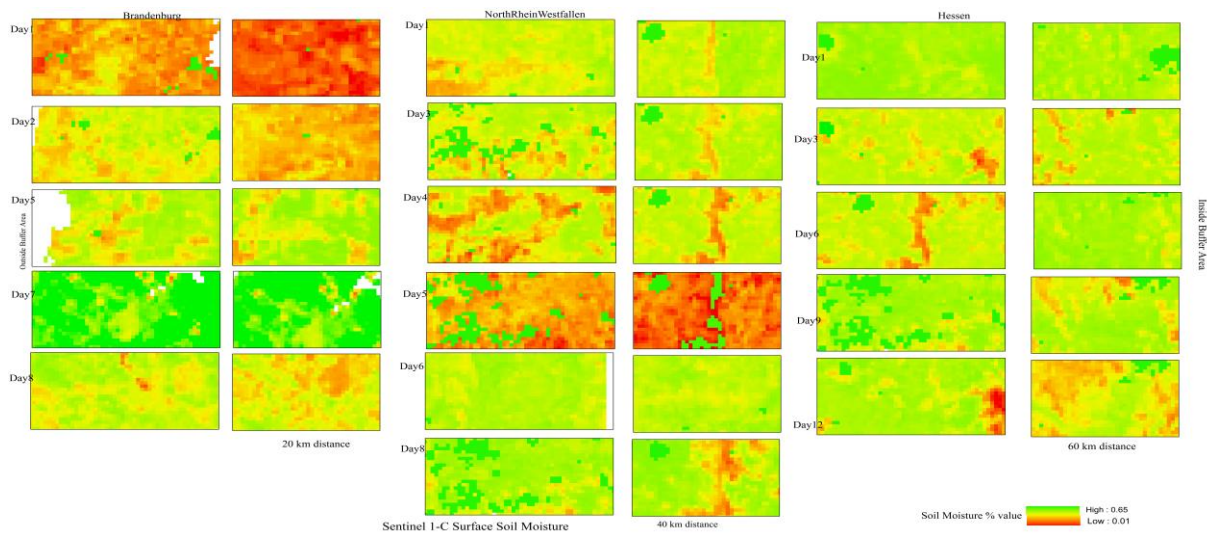


Figure 5: The soil moisture observation of Sentinel 1-C with 1 km resolution depending upon best early availability of footprint.

Sentinel 1-C data windthrow region: Soil moisture pixel grid of raster on first day on Brandenburg especially on Eberswalde region shows, the intermittent footprint of 5 days on soil map shows that day having lower soil moisture whereas day 2 and day 5 displays gradual increase of moisture content on pixel inside reaching peak on 7th day after the storm and drop their wetness resulting on less soil moisture values. In NRW day 5th shows less pixel values from soil moisture grid where day 1 to day 3 soil moisture value increase and 4th day shows darker brown pixel showing dropping in number followed by day 6 similar pixel on inside and outside region higher pixel count on days. In Hessen Sentinel-1 C surface soil moisture seems higher whereas on day 6 on storm day pixel values were higher on the windthrow region where day shows higher pixel values followed by inside region and later pixel values gradually reduce on day 12 in outer buffer zone and spotted lower on inside buffer zone after the post storm.

Pixel values analysis: For the storm event, combined data used in Soil Surface Moisture (SSM) estimating from spatiotemporal radar sensor by temporal filtering. The local soil moisture signal is highly co-related with the regional level which is outer side of buffer zone. When this holds true than individual storm zones were subjected to persistent regional soil moisture pattern showing related wetness dynamics. For accurate analysis resolution both satellite data don't necessary record in much differentiation of soil moisture events where both case soil moisture is validated despite of spatial

discrepancy. During image analysis, pixel from the darker range to lighter range shows change analysis, where a positive change analysis is noted when pixel becoming lighter and brighter when soil moisture values shift from lower state to higher state whereas negative changes analysis identifies pixel that becomes thinner or dimmer than initial state brightness.

Additionally, difference in the instrument sensor from satellites within separate time frame along with independent date and time, through non-identical atmospheric conditions as well as dominant weather conditions can affect atmospheric transmission and scattering.

Weather Maps Visualization According to Meteorological Stations

Precipitation data visualization series on various Federal States located near windthrow event.

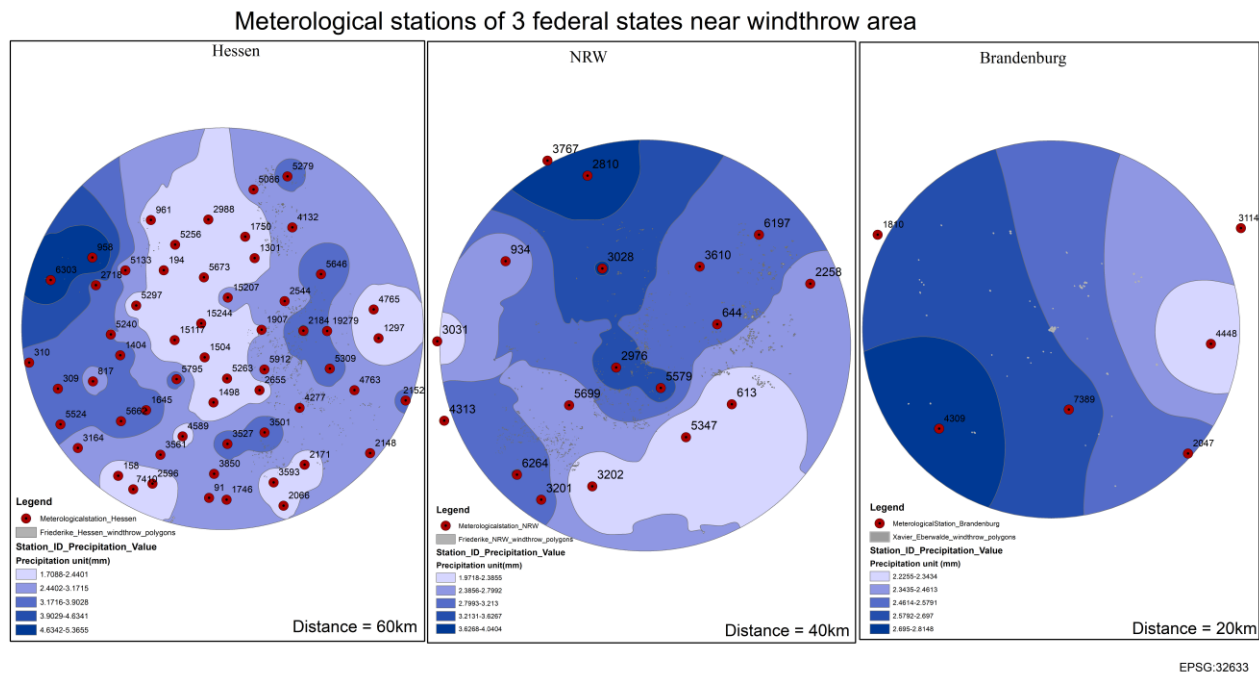


Figure 6: Weather stations of three study region

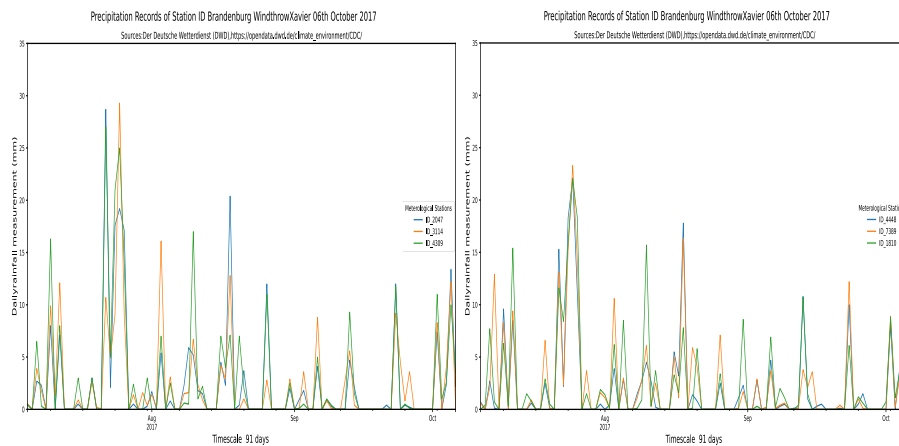
Figures 6 show rainfall recorded from various locations in the study area in 12-13 weeks during the pre- and post-storm event. The precipitation recorded values shown in the map are all the mean values of total rainfall that occurred in millimeters (mml) of each station. The federal state Hessen has a higher amount of liquid precipitation 5.35 mm, compared to NRW 4.04 mm and Brandenburg 2.8148 mm. Similarly lowest mean rainfall record was 1.708,1.9718, 2.22 was calculated in Hessen, NRW and BB. Therefore, Hessen's infiltration seems higher compared to the other two states. For the best data interpolation, Inverse Distance Weighting (IDW) was used to determine the average rainfall during the event. The approach used on IDW was to classify the mean rainfall from the meteorological station. It can be observed from the map that the Hessen region has significant rainfall records, compared with NRW and Brandenburg. The result shows most polygons were laying on the higher

rainfall area with some exceptions, which show anomaly behavior on the soil moisture on the ground truth time series visualization of precipitation datasets.

Rainfall data visualization on three different states

Precipitation Brandenburg Region

The graphical representation of time series daily rainfall records from Meteorological Station Brandenburg Region on Windthrow Location.

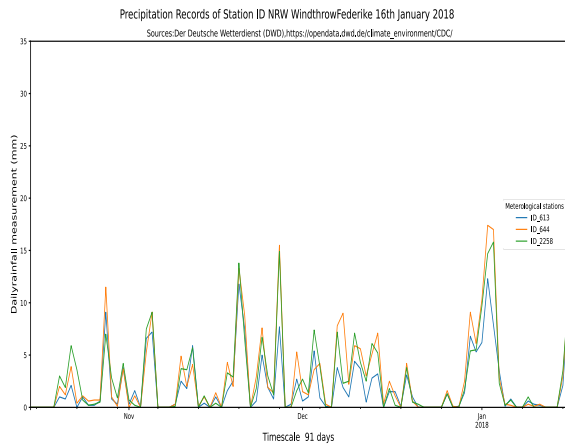


Graph 1 Precipitation graph from meteorological stations near the windthrow location Brandenburg

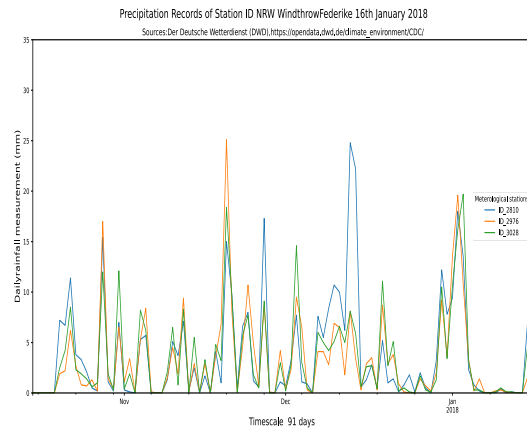
On Graph 1, meteorological stations located on the Brandenburg area near windthrow location are found to be six, where 12-13 weeks of daily rainfall events are records during the event. During the observation on the month of August all the weather stations recorded higher amount of rainfall especially weather station 3114 ,73,2047 shows much downpour on Brandenburg region.

However on the storm event on October downpour was less compared in earlier month.

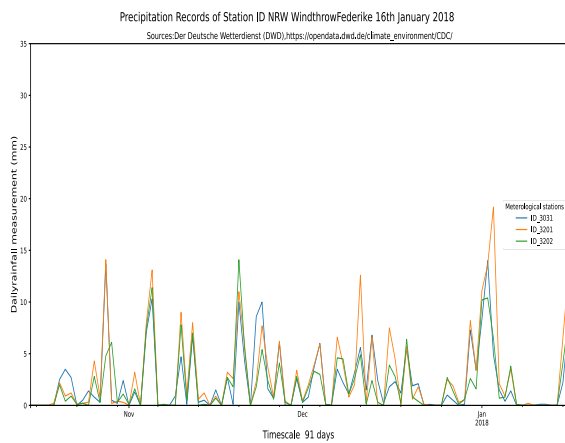
Precipitation of NRW Region (Total 18 Station) nearby windthrow location



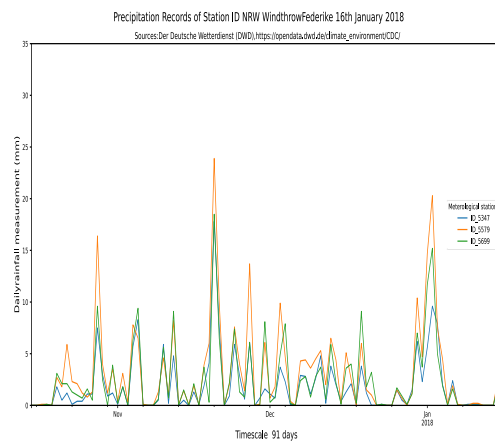
Graph 3 Precipitation records of NRW windthrow with station 613,644,2258



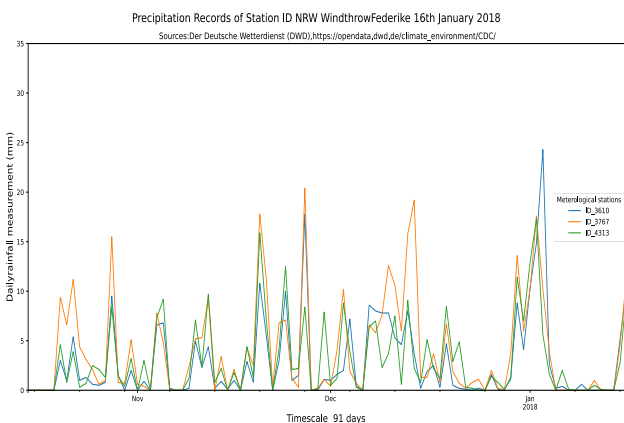
Graph 4 Precipitation records of NRW near windthrow with 2810,2916,3028



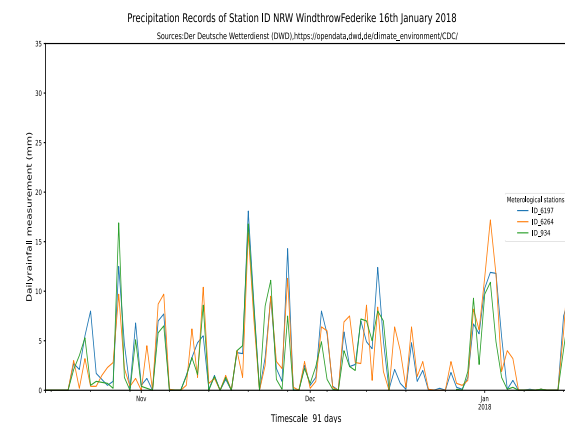
Graph 5 Precipitation records of NRW windthrow with station 3031,3201,3202



Graph 6 Precipitation records of NRW windthrow with stat 5347,5579,5699



Graph 7 Precipitation records of NRW windthrow with station 3610,3767,4313



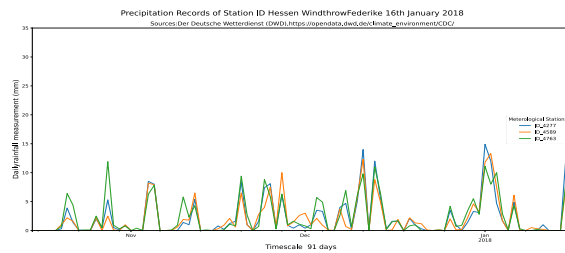
Graph 8 Precipitation records of NRW windthrow with stat 6197,6264,934

The graphical description of the time series of daily rainfall from 15th October 2017 till 16th January 2018 on the Federal state of NRW on windthrow location

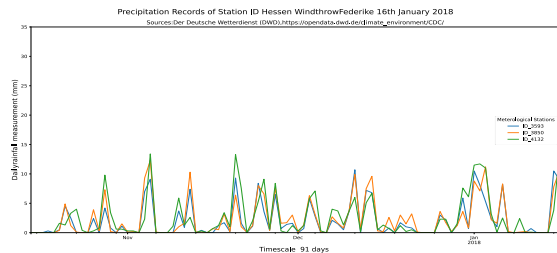
The graph 3-8 provides precipitation records weather stations of NRW, for 12-13 weeks temporal precipitation during storm Frederika. The observations that different weather stations show different rainfall records which are related to soil moisture values on the wind location. The station Id 2810 and 2916 documented higher amount of rainfall on the month of December followed by 5579 and 5699. The precipitation was higher on the station 3610 on the month of January. During storm event weather some stations like 2258,3028,5579,3767 recorded higher water precipitation which is directly related to soil moisture increase on the windthrow polygons.

Precipitation of Hessen Region (Total 56 Station) nearby windthrow location -1

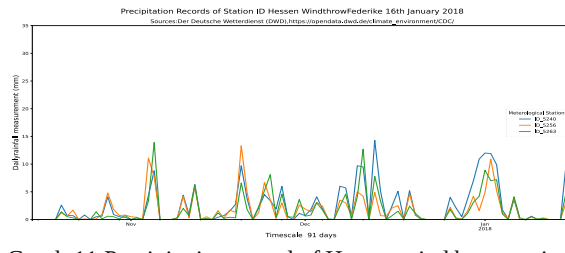
Figure: The graphical representation of the Daily Rainfall Records from 15th October 2017 till wind throw Friederike storm 2018 January 16th on Federal State Hessen



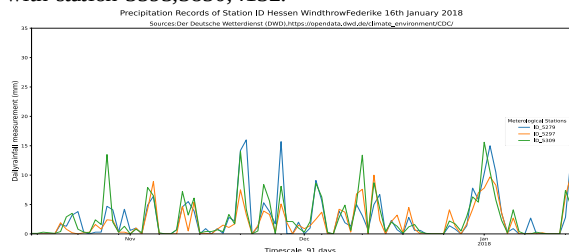
Graph 9 Precipitation record of Hessen windthrow region with station 4277,4589,4763.



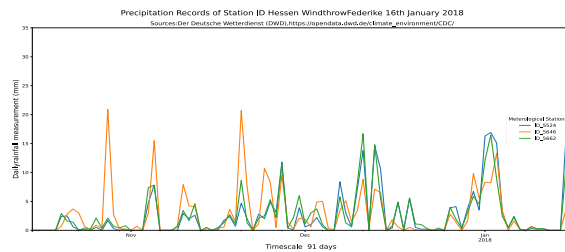
Graph 10 Precipitation record of Hessen near windthrow with station 3593,3850,4132.



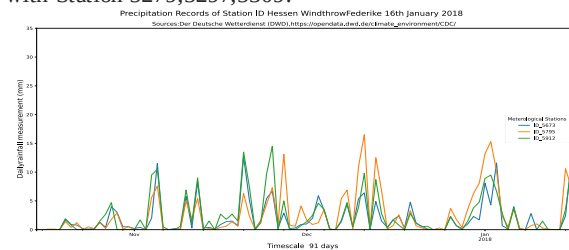
Graph 11 Precipitation record of Hessen windthrow region with station 5240,5256,5263.



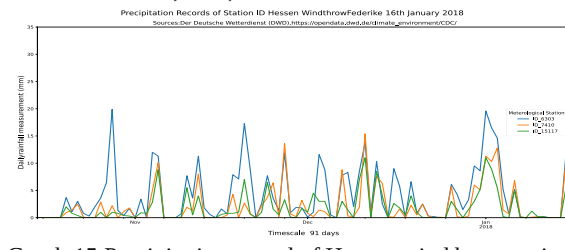
Graph 12 Precipitation record of Hessen near windthrow with Station 5279,5297,5309.



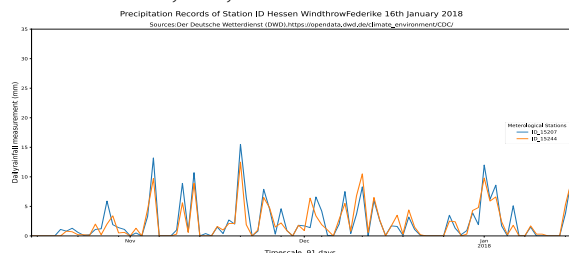
Graph 13 Precipitation record of Hessen windthrow region with station 5524,5646,5662.



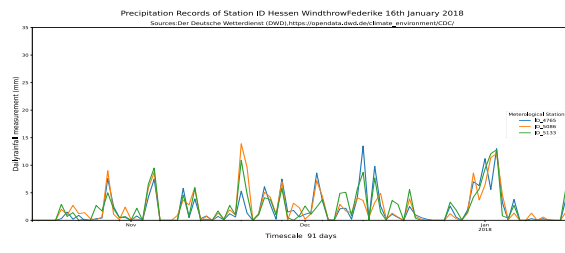
Graph 14 Precipitation record of Hessen near windthrow with station 5673,5795,5912.



Graph 15 Precipitation record of Hessen windthrow region with station 6303,7401,1517

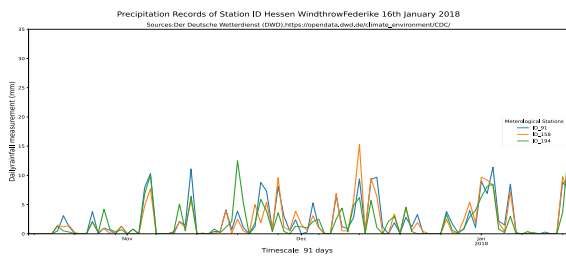


Graph 16 Precipitation record of Hessen windthrow region with station 1507,15244

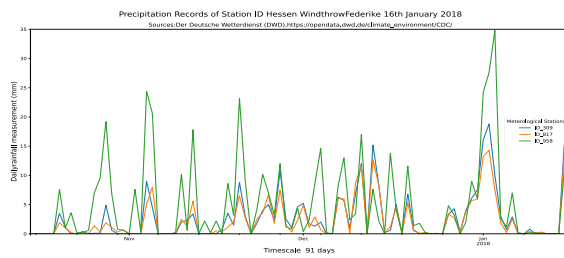


Graph 17 Precipitation record of Hessen windthrow region with station 4765,5086,5133

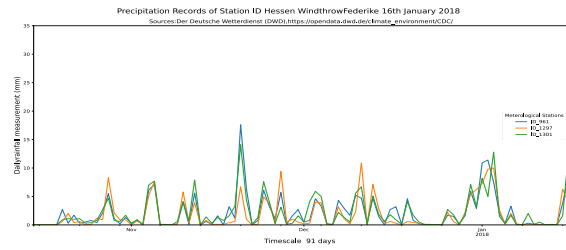
Precipitation of Hessen Region (Total 56 Station) nearby windthrow location continuous -2



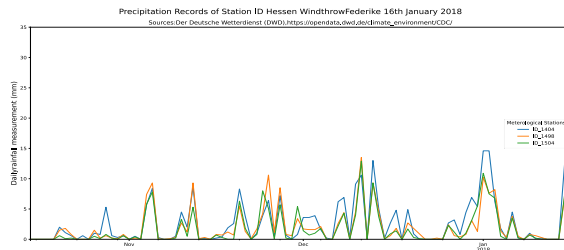
Graph 18 Precipitation record of Hessen windthrow with station 91,158,194.



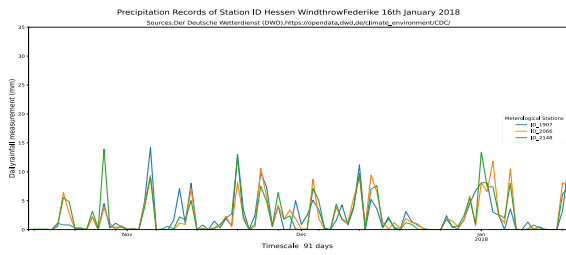
Graph 19 Precipitation record of Hessen windthrow station 309,817,958.



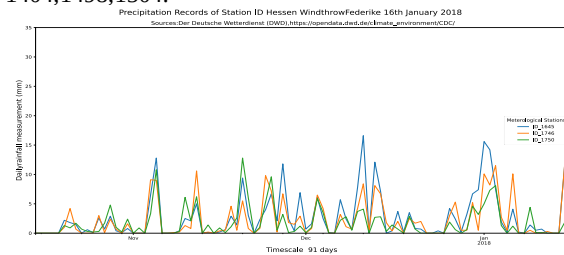
Graph 20 Precipitation record of Hessen windthrow with station 961,1297,1301.



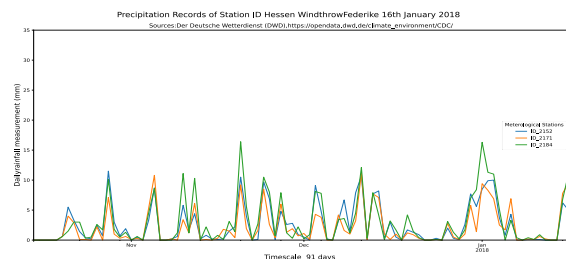
Graph 21 Precipitation record of Hessen windthrow station 1404,1498,1504.



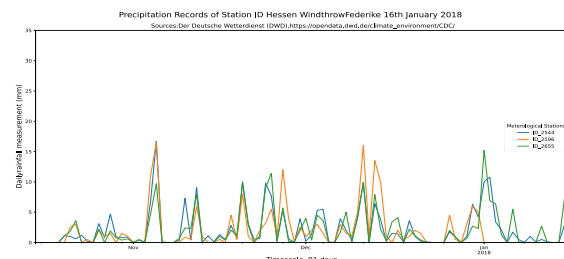
Graph 22 Precipitation record of Hessen windthrow 1907,2056,2148.



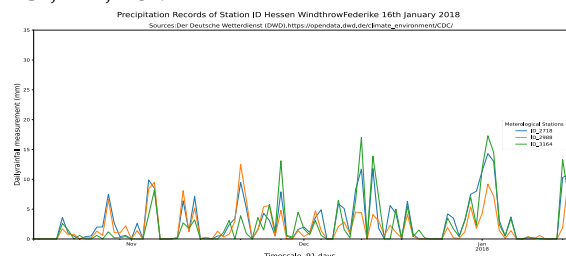
Graph 23 Precipitation record of Hessen windthrow 1645,1746,1750.



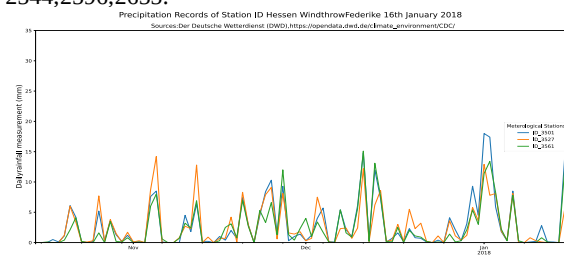
Graph 24 Precipitation record of Hessen windthrow 2152,2171,2184.



Graph 25 Precipitation record of Hessen windthrow 2544,2596,2655.



Graph 26 Precipitation record of Hessen windthrow with station 2718,2988,3164.

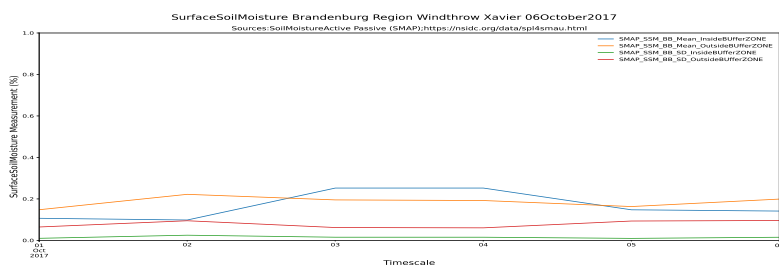


Graph 27 Precipitation record of Hessen windthrow with station 3501,3527,356.

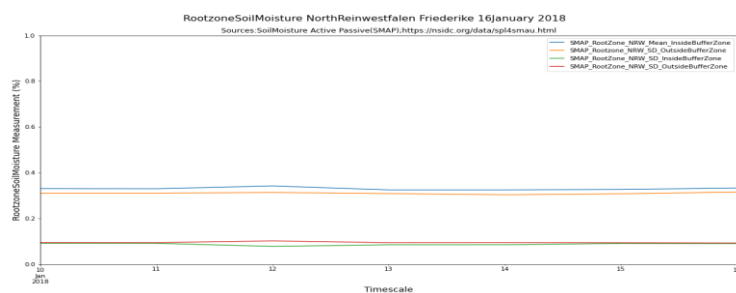
The graphical representation of time series of daily rainfall events from 15th October to January 16th during the windthrow Friederike storm on Hessen

The graphs 9-27 provide precipitation records of weather stations of Hessen for 12-13 weeks temporal precipitation during storm on this region. It can be observed that different weather stations exhibit rainfall records which are related to soil moisture values on the wind location of this region. The station Id 2810 and 2916 documented higher amount of rainfall on the month of December followed by 5579 and 5699. The precipitation was higher at the station 3610 in the month of January. During storm event weather stations with station IDs 2258,3028,5579,3767 recorded higher water precipitation which is directly related to soil moisture increase on the windthrow polygons.

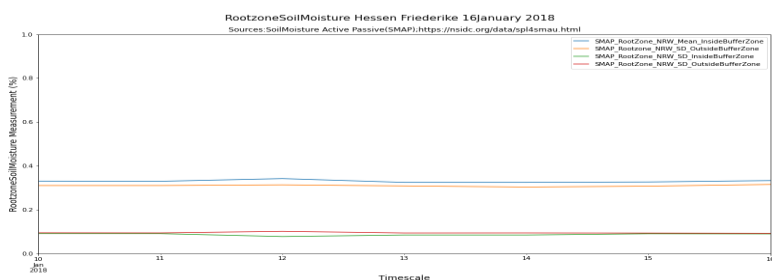
Surface Soil Moisture visualization concerning pixel values windthrow polygons.



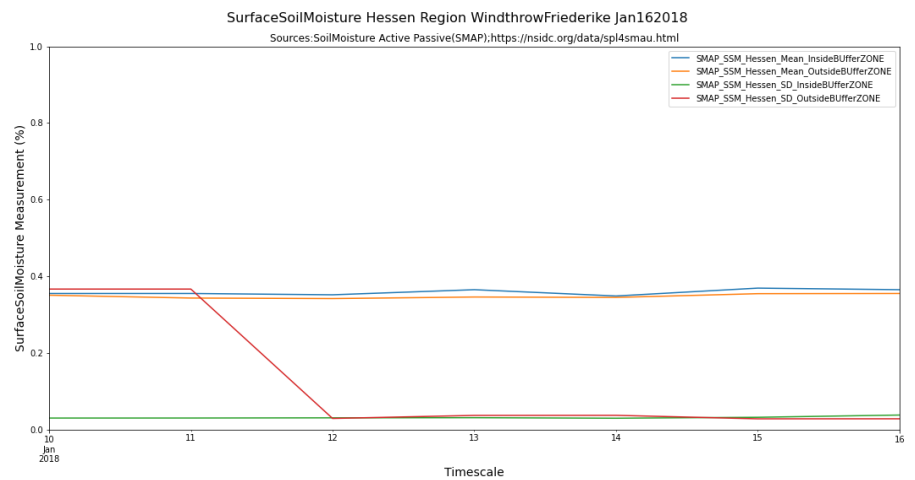
Rootzone soil Moisture of Time Series Footprint of the Satellite SMAP with 9 Km spatial Resolution in Location



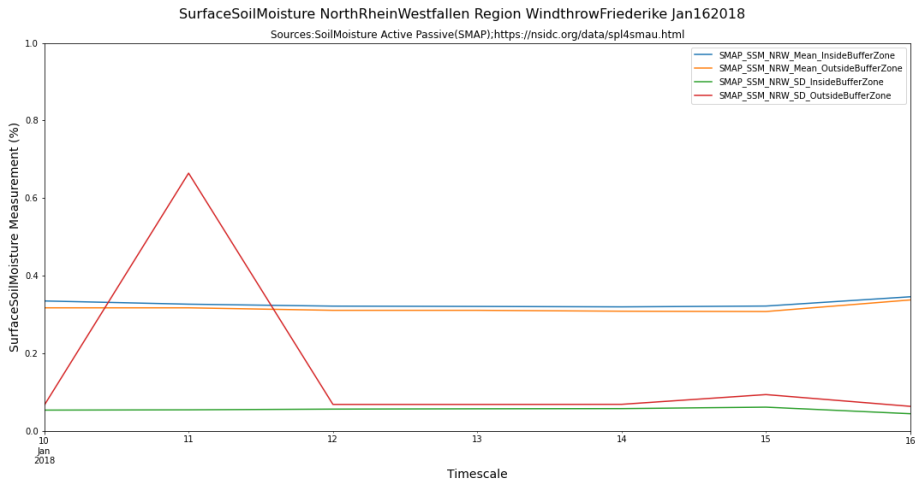
Rootzone soil Moisture of Time Series Footprint of the Satellite SMAP with 9 Km spatial Resolution in Location



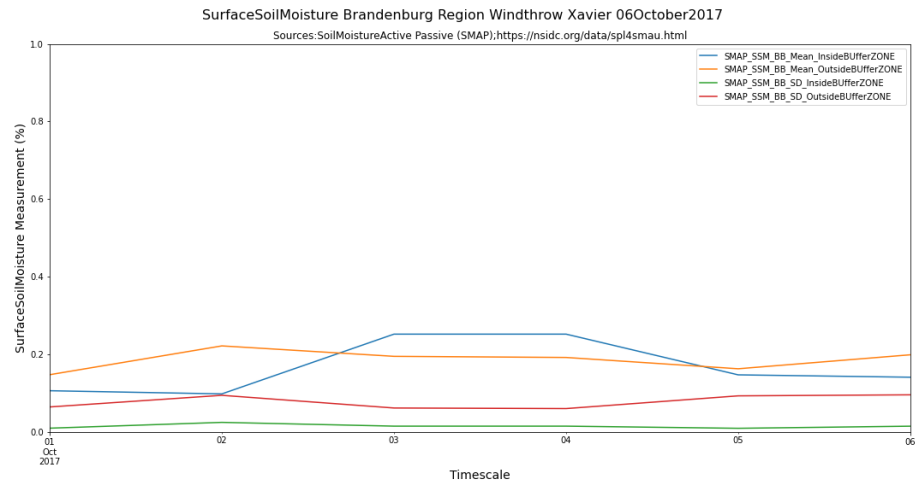
Rootzone soil Moisture of Time Series Footprint of the Satellite SMAP with 9 Km spatial Resolution in Location



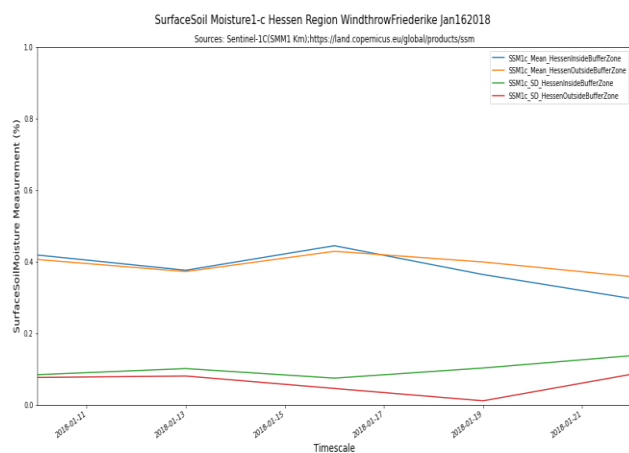
Graph 27: Surface soil moisture of SMAP on location Hessen near windthrow area



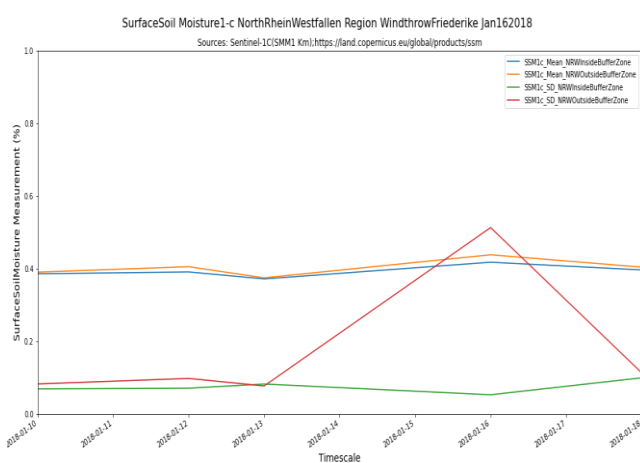
Graph 28: Surface soil moisture of SMAP on location NRW near windthrow patches



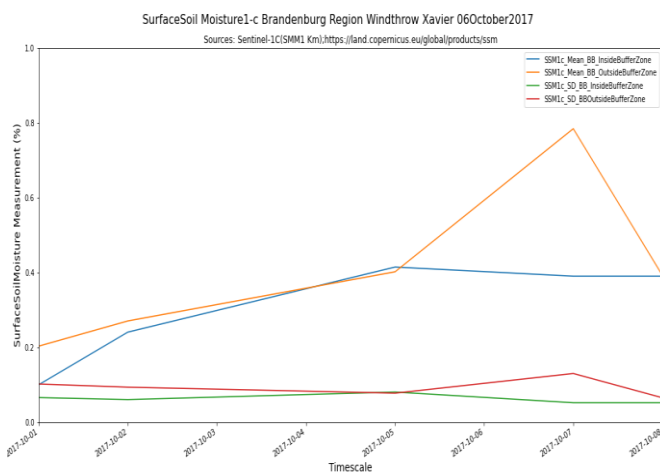
Graph 29: Surface soil moisture of SMAP on Brandenburg Region near windthrow region



Graph 30: Surface soil moisture time series footprint of Sentinel 1- C 1 km spatial resolution in location Hessen



Graph 31: Surface soil moisture time series footprint of Sentinel 1- C 1 km spatial resolution in location North Rhein-Westfalen



Graph 32: Surface soil moisture time series footprint of Sentinel 1- C 1 km spatial resolution in location Brandenburg

Spatial and temporal variability of three different datasets

The graph 27,28,29 related to soil surface moisture of SMAP satellite of 9 km resolution sensor on the 6-day series with respect to precipitation events. The SMAP mean soil moisture values inside and outer buffer seem uniform on Frederika storm in 2018 whereas soil moisture standard deviation (0.4 to 0.2) changes from to inner zone .While in NRW soil moisture values shows uniform and slight increase in inside zone compared to outer region possible difference 0.03% in moisture content, standard deviation (Sd) values from outside shows increased in beginning on first 2 days and later sharp drops from day 3 with slight increase on the soil values during the storm time. The surface moisture on Brandenburg shows fluctuation where mean moisture contents in early two days shows stable, sharp increase on 3rd day and remains steady for 2 days and finally drops on the storm time, where deviation of moisture does not show much in inner region.

The graph 30, 31, 32 related with surface soil moisture of Sentinel 1-C with 1 km spatial resolution of 5 cm layer on different locations with first footprint available on study location. The mean soil moisture values on Hessen region shows about 0.4 % on inside and outside buffer region in beginning whereas increase almost 0.01 on wind events and sharp decrease after the event with much deviation on the values around 0.1 % inside the polygons region whereas NRW outer soil moisture was found to 0.4 % slight increase on the period of storm phase and little drop after the event .Around 0.5% Sd was observed on the outer buffer zone showing clearly different in moisture contents whereas outside buffer polygon region is much higher of changes of moisture. On Brandenburg region soil moisture shows increase trend in inside region from 0.1% to 0.4% in windthrow area whereas outer zone 0.2% to 0.65% during the storm. The Sd on both regions were less than 0.15 % and outer region after storm higher deviation values.

The graph 33, 34 ,35 related to rootzone soil moisture on SMAP Rootzone soil moisture on six days regular footprint availability on three locations. The inside and outside mean value

The wetness of underground soil moisture seems uniform where soil moisture inside buffer region seems always higher despite of area cover on Frederika storm event however the soil moisture before Xavier storm event seems higher on Brandenburg area on the year 2017.

Discussion

On the final analysis, two windthrow were observed, year 2017 and 2018. The temporal variation of rain event (12-13) weeks was observed. The spatial and temporal comparison of soil moisture raster pixel from satellite data were observed with precipitation data along with surface soil moisture and rootzone moisture especially with rain event. Notable soil moisture wetness was observed among the three-study region. The result shows ,soil moisture content increasing after precipitation in the Brandenburg region shown from weather station , which seems consistent with temporal alternation

with rainfall pattern. The water graph shows much difference of higher spikes when high humidity on the wind phase of 2017 as also in 2018. The rainfall record on the nearby station shows higher amount of water downpour on Hessen as compared with NRW and Brandenburg. It can be assumed that mean soil moisture on observation were closely associated with number of clusters in different study location. When carrying raster grid analysis approaches like zonal statistics, point sampling during pixel extraction from datasets shows that variation of soil moisture distribution on the individual polygons. The soil moisture distribution of windthrow cluster with spatial resolution Sentinel 1-C data observed on polygons shows much clear soil moisture content possibly because of higher resolution whereas from the SMAP soil moisture datasets provide continuous moisture distribution on the windthrow clusters but not much information on individual polygons. Other assumption regarding moisture, In all federal states where wind event occur, soil and its moisture contain totally depend on upon the external features like humidity and precipitation as well as other external features (González-Zamora, Sánchez, Martínez-Fernández, & Wagner, 2016). For the same event of federal state NRW and Hessen, the possible reason on different soil moisture value seems different because of texture slope or local geography itself. . In remote sensing of microwave satellite, pixel values vary, because of backscatter response, a main constituent's agent accountable for variation for surface moisture. When backscatter is higher than accessibility of footprint is greater whereas lower scattering responsible for less moisture obtainability on backscatter where intervention from vegetation and soil conditions (Petropoulos, Ireland, & Barrett, 2015). Since, bare soil surfaces are quite sensitive to moisture due to the dielectric constant on increasing water content (Ulaby et al., 1982).

In microwave remote sensing, the active sensor i.e., Scatterometers, transmits electromagnetic rays and measures the earth's surface energy, whereas in radiometers, energy is targeted by itself source relating to emitting layer and emissivity of the surface, which are closely related to the measurement of Kirchhoff's law, where earth surface and reflectivity are measured however the signal varies depending on the characteristics and mode of sensor. This shows the essential aspects of the soil moisture to go further into the deep or deep soil layer (Baldwin, Manfreda, Lin, & Smithwick, 2019).

While considering raster pixel appearance from the data containing different in values, the resulting pixel were changing from darker, lighter, dampen and mixed depending upon the moisture contents. The available water content also changes when the temperature changes (environmental influence). The fluctuation of the graph was spotted in every figure related to soil moisture and precipitation possible causes of the humidity, dryness, and temperature. The water content on the storm areas depends upon the water movement where deposition is seen during the precipitation. The temperature changes on the precipitation, i.e. winter month, especially on wind throw Federike the soil water on the surface as well as root zone likely to be saturated or freezes where fluctuation on the graph is observed. However, the soil is wet again due to the precipitation and moisture level rises with soil

moisture percentage. This shows in gaps and the moderate volatile nature of water. The numerous altering patterning of graph shows changing wetness on moisture and varying in the precipitation event.

Rainfall Data Visualization and Preprocessing

The daily rainfall data from individual stations extracted from the DWD. The numerous information related to weather record is linked with attributes from the time of establishment till their operation. During the data acquisition process from data records. The rainfall events were noted regularly and tabulated in the CSV format with date, rainfall amount, latitude, longitude, height, with station number. The nearby located stations were georeferenced with nearby coordinate in operation location assigned with UTM zone with provided parameters mentioning in the csv for shaping the individual file with the attribute along with the meta data information from DWD. For precision and accuracy of rainfall stations, rain recorded from weather station of each federal state of windthrow damages patches to the wind event. The cleaning process of some missing values has not defined a precise methodology however mean or zero value was added where no information was found at the data cleaning. The approach used for missing values in the rainfall datasets is Heuristic-based, making a reasonable guess based on the knowledge of the underlying station. In comparison, considering the middle part of data, i.e., means. The mean position values are considered the safest value for changing because of their nature; filling the missing data with mean values doesn't hamper the result. During the visualization of precipitation from the weather station, the heterogenous message was observed where several stations had no precipitation records value. During visualization, NA records were replaced or omitted, while summarizing the data summary of each station was tabulated, and visualization was performed. Regular rainfall measured record from the DWD as ground truth data for finding soil humidity. After data cleaning, the mean value of rainfall of the station was calculated for generating precipitation maps near the underlying polygons near the area. For interpolation, Inverse Distance Weighting (IDW) was used for figuring the average rainfall during the event. The approach used on IDW was classify the average rain from the meteorological station. It can be observed from the map that Hessen region has major rainfall records whereas NRW and BB. The result shows major polygons were laying on the higher rainfall area with some exceptions which show anomaly behavior on the soil moisture on the ground truth data.

5.4 Footprint gaps of Sentinel 1- C

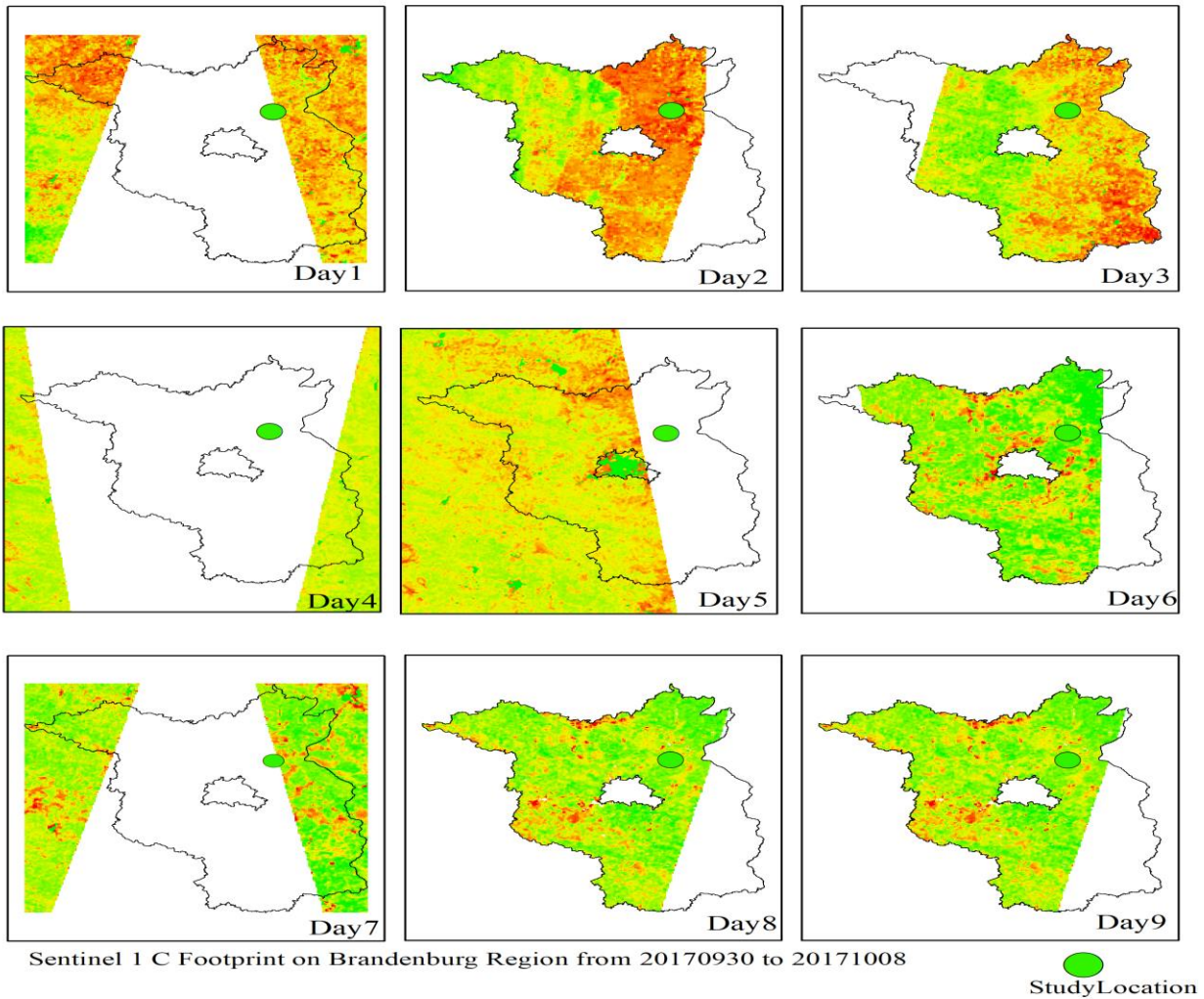


Figure 21 Soil moisture footprint of Sentinel 1-C 1 km resolution in single frame

The figure describes an example of temporal soil moisture footprint availability in the federal state Brandenburg. On the acquisition process for analysis, suitability, and continuity of Sentinel 1- C soil moisture data, footprint data seems to be available but not in desired way, strap shape of Sentinel 1- C brought some challenges during the analysis and questions about its suitability showing no footprint, partial data, half footprint data, as well as data behind the study location. In other study areas a similar pattern was found. However, SMAP data were continuous and regular available despite of low resolution than Sentinel 1-C. The final visualization of the satellite image (SMAP) with lower spatial resolution along with RZSM to the satellite data images with same resolution surface soil moisture with regarding the higher spatial resolution Sentinel 1-C SSM 1 km spatial resolution shows the moisture where post windthrow shows abrupt changes of soil moisture on 6 days event in federal states on Brandenburg as well as NRW and Hessen in two different windstorm events together with temporal frame. For the in-depth analysis concerning soil moisture windthrow cluster longer time frame and in-situ data plays vital role. The regular raster image acquisitions during the visualization, and observation from the different windthrow locations detected the changes in the

clear soil moisture changes into dry and wet, showing the foresight of meteorological effects i.e., windthrow as well as precipitation. It can simply generalize that a rapid increase in the soil moisture on the topography of the soil can create concern over sudden natural events such as windstorm, or floods which can be helpful for farmers and forester could be the best way to predict and minimize the effect and loss.

Change Analysis: The Observed maps during the event have both spatial and temporal variability from the meteorological event reflecting soil texture variability which could be an activity of forest harvesting of study region. From the Satellite data, a higher amount of soil moisture conditions was noticed at the time of the storm event while in Brandenburg post windthrow effect was also present. During the Xavier (2017) storm the (6 days) as well as Frederika (2018) time frame. The footprint data inside and outside the buffer zone on the Brandenburg Region shows soil moisture content in contrast on the following dates are higher whereas moisture on other days the soil moisture was low.

Scopes, and Limitations

This research is based upon suitability and feasibility of remote sensing data. During the research progress, some limitations were observed in the methodology. Firstly, Sentinel 1 – C data were not consistent on the regular basis during the wind event which created difficulty on comparison of datasets. Secondly, windthrow events were observed on distinct location on independent time frame with special geographical attributes. Afterwards windstorm had damaged affected areas in unequal spatial pattern where polygons were scattered very diversely on the study locations. At the time of visualization of raster grid of soil moisture datasets, before event and on event for 6 days so much temporal precipitation event data, post storm phase were not listed after the storm phase. Lastly some windthrow polygons were very small in area during buffering while extracting soil moisture values are main underlying condition of this research.

Conclusion

In this research, footprint of data availability and suitability was performed on the three different soil moisture value containing weather datasets in the basis on spatial and temporal content of soil moisture data where the final time series result was obtained. It can be concluded that during the visualizations, the temporal variations of mean soil moisture does not show consistency with the rainfall activity due to lower number of soil moisture percentage value. The windthrow polygons also show lag effect following the rainfall due to buffering influences on the topography. Moreover, soil water in upper surface area, root zone indicating difference on the patches of study region.

The footprint availability of remote sensing data and adequate enclosing of their study area with the adequate coverage platform with sufficient imagery accessibility as well as locating sufficient coverage of one's area of interest can be difficult, particularly if the study topic is large-scale or global. The ability to quickly view footprint enclosure areas by platform indicates both the depth of imagery that is currently available as well as where data gaps exist. For precise remodeling and shifting, detection needs a long range of events for soil moisture observation. For a short-range change approach influencing factor like soil roughness, tree height, vegetation as well as water content, the radar backscatter on temporal changes on the surface can be useful only considering multiple factors keeping statics on the region area, dominated attributes like water bodies, urban areas, dense forest making radar sensitivity much weak and poor during data acquisitions of SSM (F.Mattia.,A.Balenzano.,G.Satalino., 2017)

Further research and test for soil moisture datasets

The research objective to find suitability and feasibility of research data for project implementation, it's known that soil moisture data has some error during acquisition. There is always a necessity for data to compare and test. There are numerous approaches where validation can be performed. The methods like Spearman Test, Pearson Correlation, Systematic bias, Root Mean Square Error (RMSE), Unbiased Root Mean Square Error as well as Normalized Standard Deviation are practiced for maximizing standardization and performance of data applicability in remote sensing field in soil moisture area. The four statistics approaches for soil moisture retrievals as well as $MD = \sum_{i=1}^N (\theta_s(i) - \theta_m(i))$ The RMSD for difference and accuracy of soil moisture can be calculated as

$$RMSD = \sqrt{(\sum_{i=1}^N (\theta_s(i) - \theta_m(i))^2) / N}$$

Where θ_s represents a satellite soil moisture retrieval ($m^3 \cdot m^{-3}$), and θ_m is in-situ moisture measurement ($m^3 \cdot m^{-3}$), where for calculation for temporal analysis with different types of samples similarly N is the incremental time steps, and its representation on daily or monthly time step.

Root mean Square: For the calculation of the bias removal, ubRMSE which characterizes random error is given by: $ubRMSE = \sqrt{((RMSD * RMSD) - (MD * MD))}$

Correlation coefficient(R) can be used for the evaluation of both satellite soil moisture data against in situ and ancillary datasets. $R = \sum_{i=1}^N (\Theta_s(i) - \mu_s)(\Theta_m(i) - \mu_m) / (N-1)\sigma_s\sigma_m$

Where μ_m is the average soil moisture during evaluation period for a footprint ($m^3 \cdot m^{-3}$) where μ_m represents the average soil moisture measurements ($m^3 \cdot m^{-3}$)

Final Remarks

Selecting satellite imagery can always be a challenging process, one cannot easily draw a conclusion stating particularly for possible options of remote sensing data for different tasks with different purposes and models. Since this research was of two different remote sensing images product for the possibility for usefulness and accessibility for footprint generation. There is always limitation on product itself, uncertainties associated downscaled model along with scales and satellite coverage, and uncertainties of in situ observations (Peng et al., 2017). It can be simply stated that it is not necessary to have always high-resolution images for every purpose and situations. However, all remote sensing data provides valuable information which is always useful for scientific studies.

Reference List

- Ahlmer, A.-K., Cavalli, M., Hansson, K., Koutsouris, A. J., Crema, S., & Kalantari, Z. (2018). Soil moisture remote-sensing applications for identification of flood-prone areas along transport infrastructure. *Environmental Earth Sciences*, 77(14).
- Arlinghaus, S. L. (2019). *Spatial thinking in environmental contexts: Maps, archives, and timelines* / Sandra L. Arlinghaus, Joseph Kerski, Ann Evans Larimore, and Matthew Naud, editors. Boca Raton, FL: CRC Press.
- Baldwin, D., Manfreda, S., Lin, H., & Smithwick, E. A. (2019). Estimating Root Zone Soil Moisture Across the Eastern United States with Passive Microwave Satellite Data and a Simple Hydrologic Model. *Remote Sensing*, 11(17), 2013.
- Balenzano, A., Mattia, F., Satalino, G., Lovergine, F. P., Palmisano, D., & Davidson, M. W. J. (2021). Dataset of Sentinel-1 surface soil moisture time series at 1 km resolution over Southern Italy. *Data in brief*, 38, 107345.
- Bauer-Marschallinger, B. (2019). *Product User Manual SSM Collection 1km Version 1*. Retrieved April 19, 2019, from TU Wein: <https://sentinel.esa.int/web/sentinel/user-guides/sentinel-1-sar>.
- Bauer-Marschallinger, B., Paulik, C., Hochstöger, S., Mistelbauer, T., Modanesi, S., Ciabatta, L., et al. (2018). Soil Moisture from Fusion of Scatterometer and SAR: Closing the Scale Gap with Temporal Filtering. *Remote Sensing*, 10(7), 1030.
- Benninga, H.-J. F., van der Velde, R., & Su, Z. (2020). Sentinel-1 soil moisture content and its uncertainty over sparsely vegetated fields. *Journal of Hydrology X*, 9, 100066.

- Cai, J. (2016). Geophysical Parameters Retrieval From Sentinel-1 Sar Data: A Case Study For High Performance Computing At EODC. In *24th High Performance Computing Symposium*. Society for Modeling and Simulation International (SCS).
- Das, N. N., Entekhabi, D., Dunbar, R. S., Chaubell, M. J., Colliander, A., Yueh, S., et al. (2019). The SMAP and Copernicus Sentinel 1A/B microwave active-passive high resolution surface soil moisture product. *Remote Sensing of Environment*, 233, 111380.
- F.Mattia.,A.Balenzano.,G.Satalino. (2017). SENTINEL-1 HIGH RESOLUTION SOIL MOISTURE. *IEEE Transactions on Geoscience and Remote Sensing*, 4. Retrieved August 06, 2022.
- González-Zamora, Á., Sánchez, N., Martínez-Fernández, J., & Wagner, W. (2016). Root-zone plant available water estimation using the SMOS-derived soil water index. *Advances in Water Resources*, 96, 339–353.
- Gross, G. (2018). A windthrow model for urban trees with application to storm “Xavier”. *Meteorologische Zeitschrift*, 27(4), 299–308.
- Hochstöger, S. (2016). TRANSITION OF TU WEIN CHNAGE DETECTION METHOD FROM ASAR TO SENTINEL1: A STATISTICAL ANALYSIS. *Computers & Geosciences*. (1027106), 81. Retrieved June 27, 2022, from <https://www.tuwien.at/bibliothek>.
- Kimerling, A. J. (2002). Predicting Data Loss and Duplication when Resampling from Equal-Angle Grids. *Cartography and Geographic Information Science*, 29(2), 111–126.
- Kreklow, J., Tetzlaff, B., Burkhard, B., & Kuhnt, G. (2020). Radar-Based Precipitation Climatology in Germany—Developments, Uncertainties and Potentials. *Atmosphere*, 11(2), 217.
- Meyer, R., Zhang, W., Kragh, S. J., Andreasen, M., Jensen, K. H., Fensholt, R., et al. (2021). *Exploring the combined use of SMAP and Sentinel-1 data for downscaling soil moisture beyond the 1 km scale*.
- Peng, J., Loew, A., Merlin, O., & Verhoest, N. E. C. (2017). A review of spatial downscaling of satellite remotely sensed soil moisture. *Reviews of Geophysics*, 55(2), 341–366.
- Petropoulos, G. P., Ireland, G., & Barrett, B. (2015). Surface soil moisture retrievals from remote sensing: Current status, products & future trends. *Physics and Chemistry of the Earth, Parts A/B/C*, 83-84, 36–56.

Reichle, R., Lannoy, G. de, Koster, R., Crow, W., & Kimball, J. (Eds.). SMAP L4 Global 3-hourly 9 km Surface and Rootzone Soil Moisture Analysis Update, Version 3: NASA National Snow and Ice Data Center DAAC.

Richard Kidd (2021). *ESA Climate Change Initiative Plus soil Moisture*. Retrieved April 16, 2021, from European Space Agency: <https://climate.esa.int/en>.

Vautard, R., van Oldenborgh, G. J., Otto, F. E. L., Yiou, P., Vries, H. de, van Meijgaard, E., et al. (2019). Human influence on European winter wind storms such as those of January 2018. *Earth System Dynamics*, 10(2), 271–286.

Annex: A Sentinel 1 – C First 5 days footprint Brandenburg

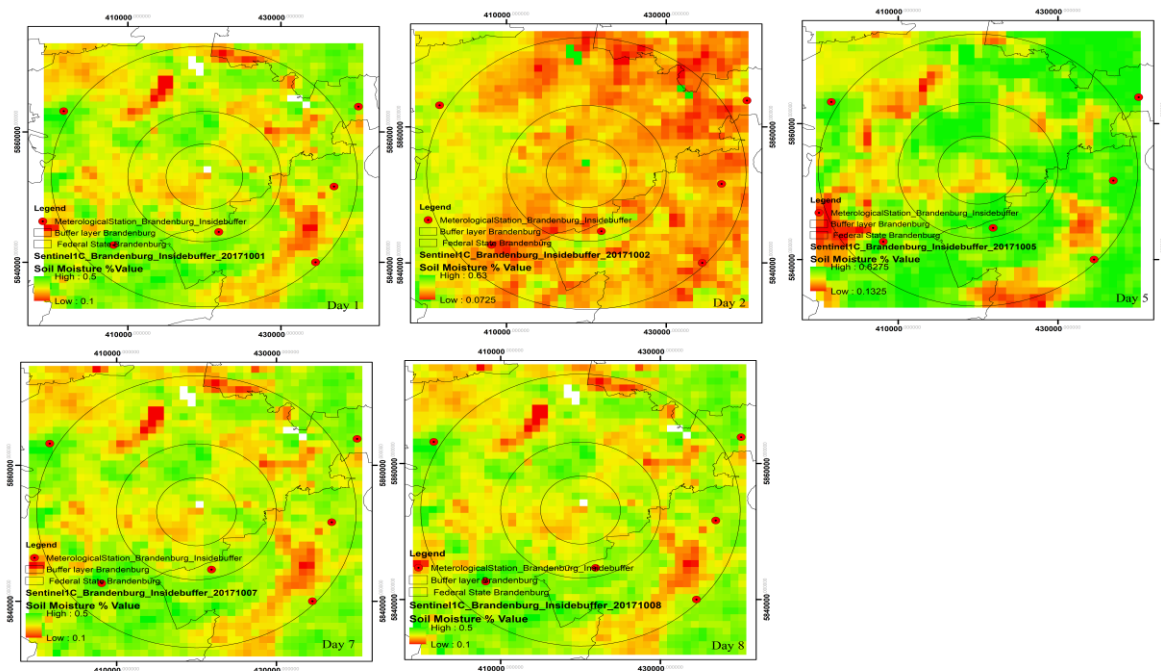
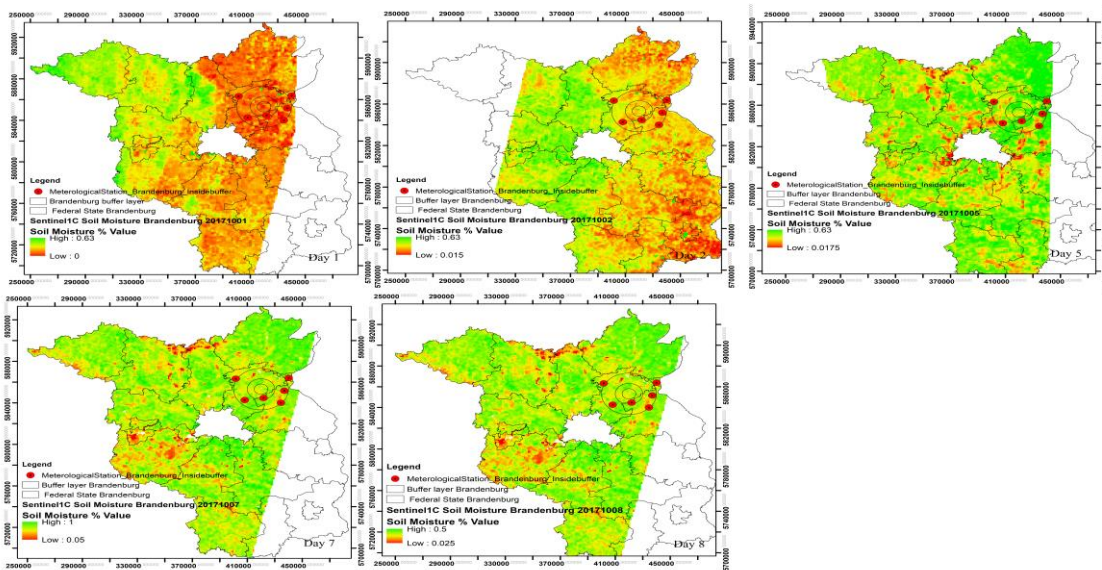


Figure Sentinel 1-C Soil Moisture Brandenburg from 2017-1001 till 2017-1008(Windthrow region)

Annex: B Sentinel 1-C Brandenburg Sentinel Footprint 5-days



1) Figure 2: Sentinel 1-C Soil Moisture Brandenburg from 2017-1001 till 2017-1008(Whole)

Annex: C Sentinel 1-C Hessen Footprint 5-days

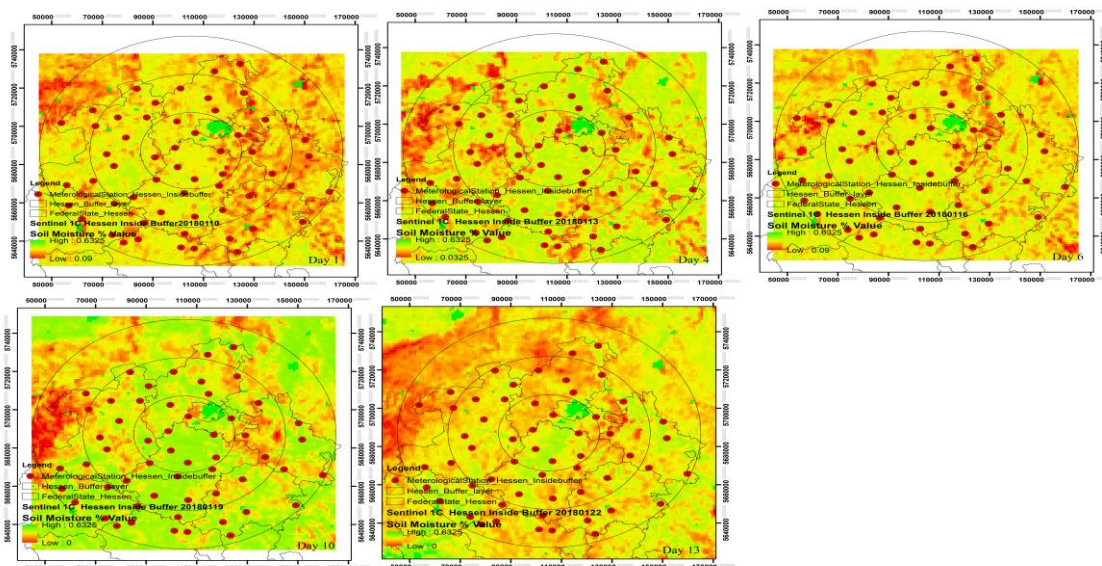


Figure 3: Sentinel 1-C Soil Moisture Hessen from 2018-01 10 till 2018-01 22(Windthrow region)

Annex D: Sentinel 1- C Hessen first 5 days footprint

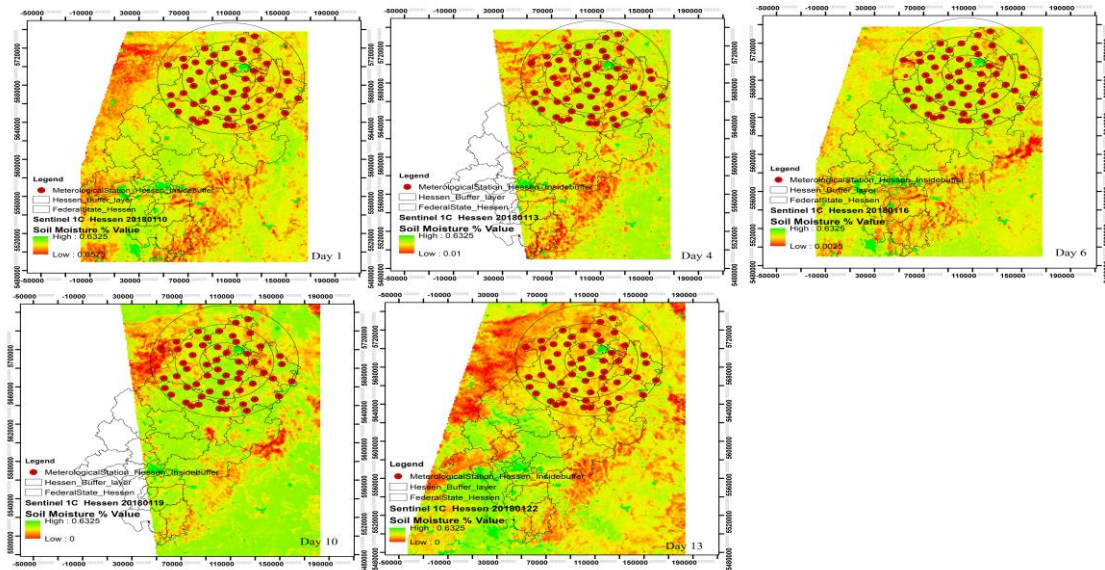


Figure 4: Sentinel 1-C Soil Moisture Hessen from 2018-01 10 till 2018-01 22(Whole region)

Annex E: Sentinel 1- C NRW first 6 days footprint

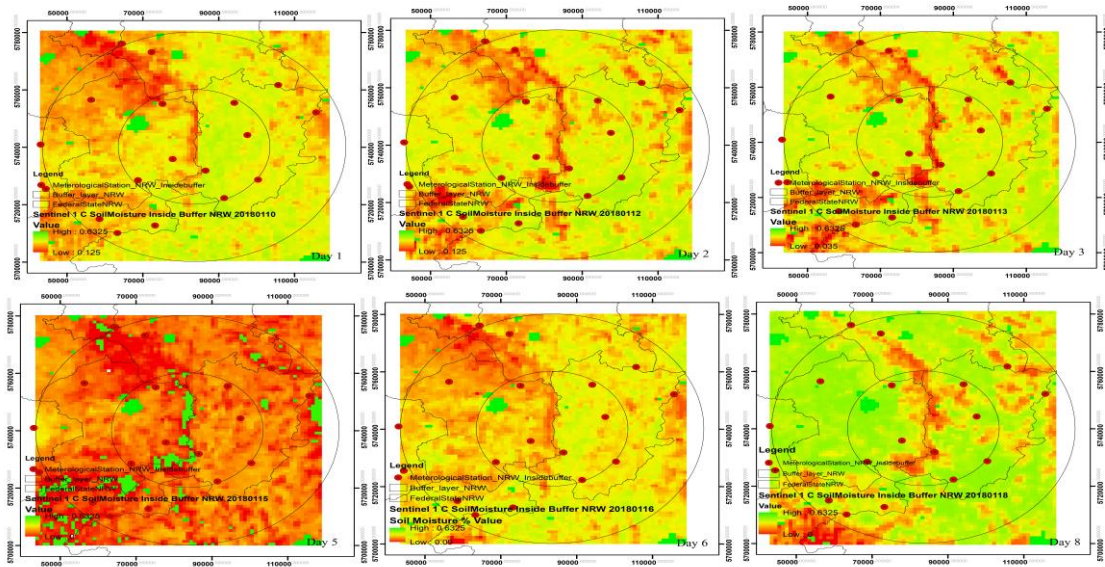


Figure 5: Sentinel 1-C Soil Moisture Hessen from 2018-01 10 till 2018-01 18(Windthrow region)

Annex F: : Sentinel 1- C NRW first 6 days footprint

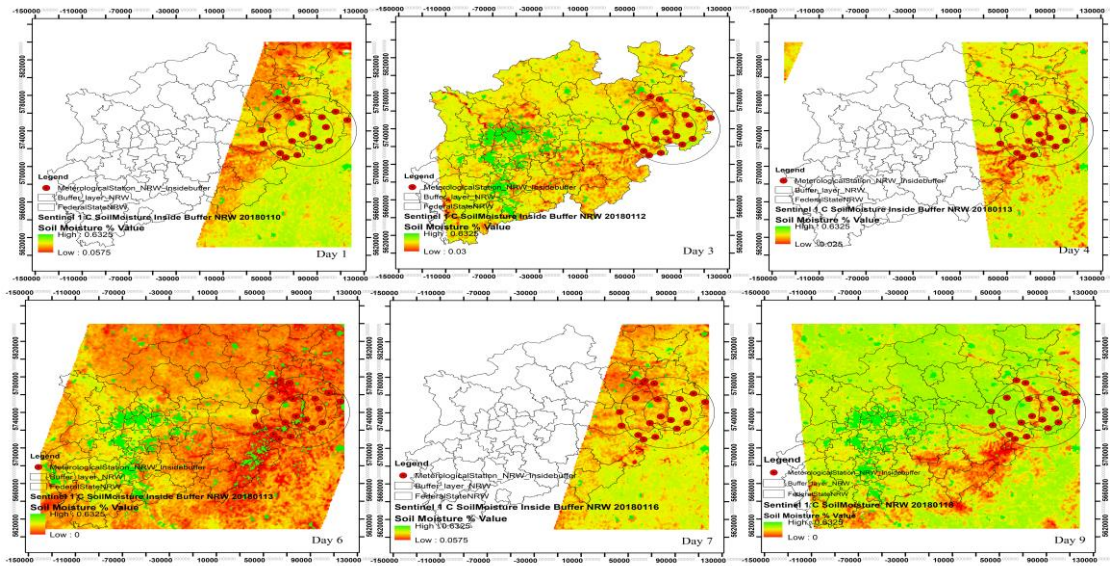


Figure 6 : Sentinel 1-C Soil Moisture Hesse from 2018-01 10 till 2018-01 18(Whole region)

Annex G: Surface Soil Moisture Footprint 6 days on Brandenburg Region:

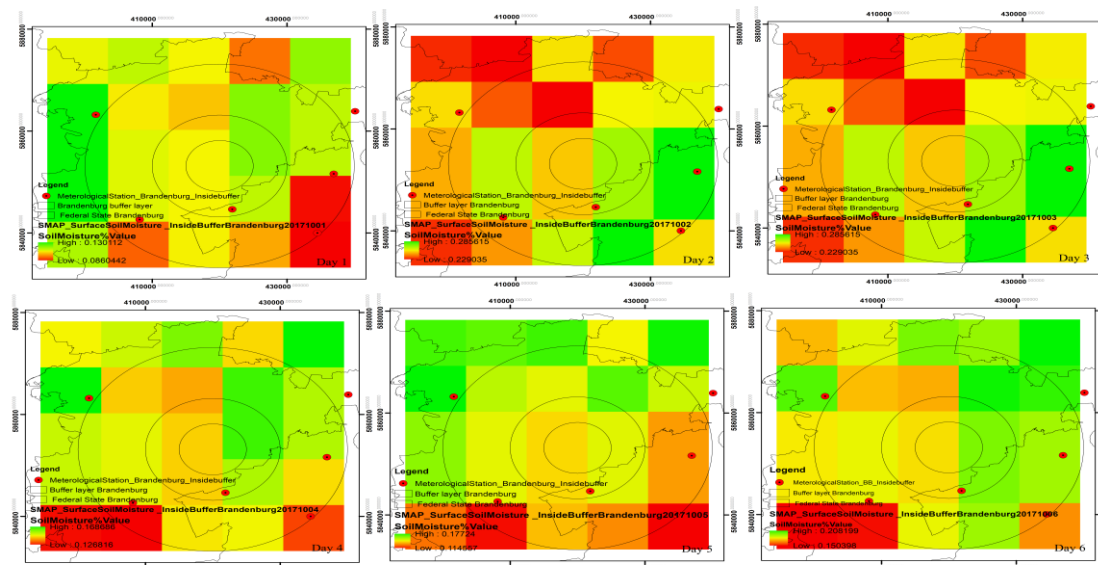


Figure 7: Surface Soil Moisture Brandenburg from 2017-1001 till 2017-1006(Windthrow region)

Annex H: Surface Soil Moisture Footprint 6 days on Brandenburg Region

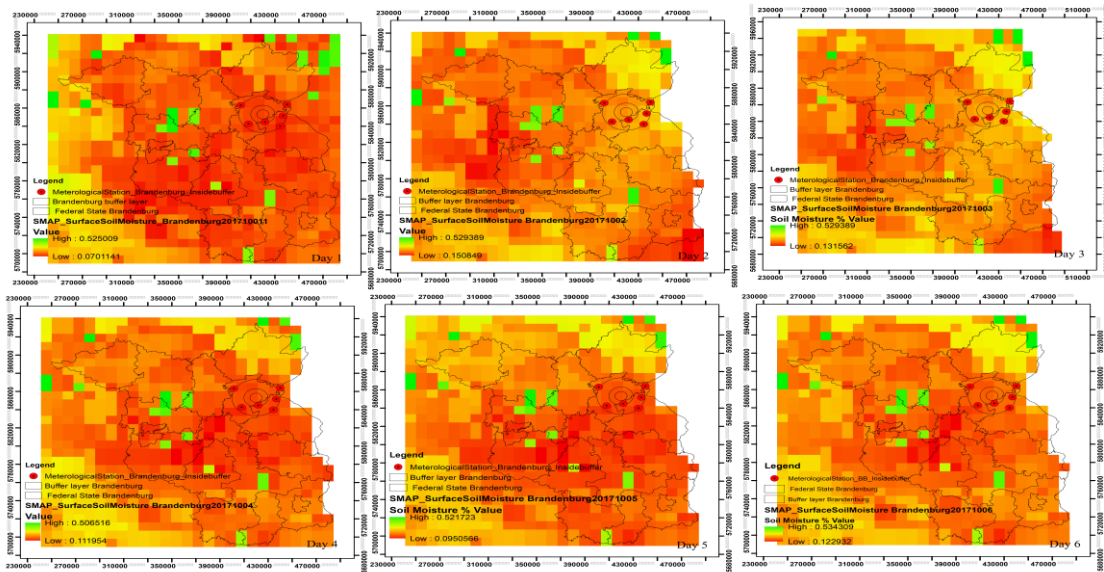


Figure 8: Surface Soil Moisture Brandenburg from 2017-1001 till 2017-1006(Whole region)

Annex I: Surface Soil Moisture on NRW 6 Day series

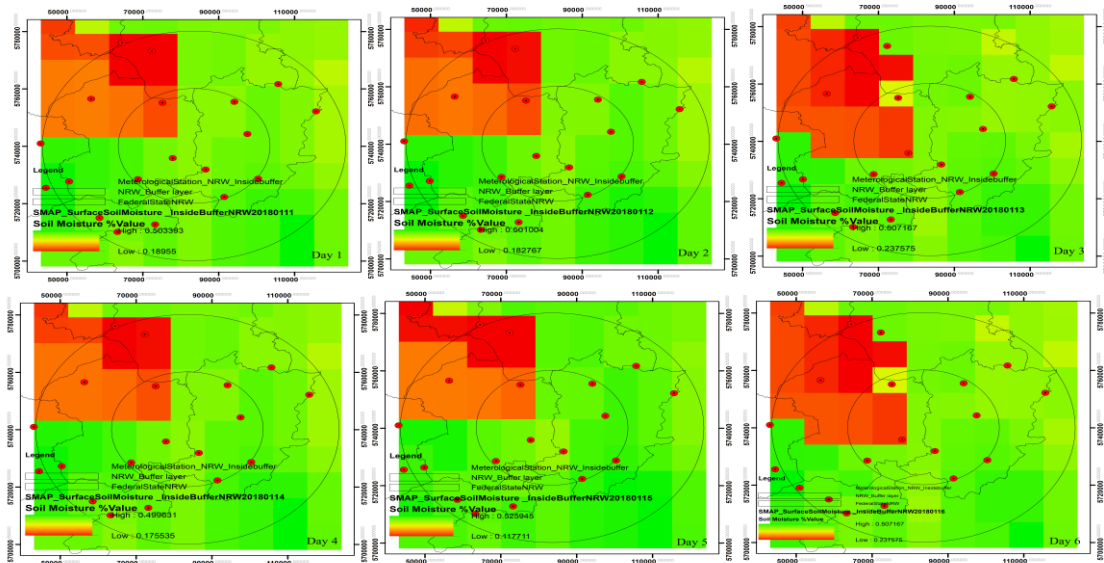


Figure 9: Surface Soil Moisture NRW from 2018-0110 till 2018-0116 (Windthrow Region)

Annex J: Surface Soil Moisture on NRW 6 Day series

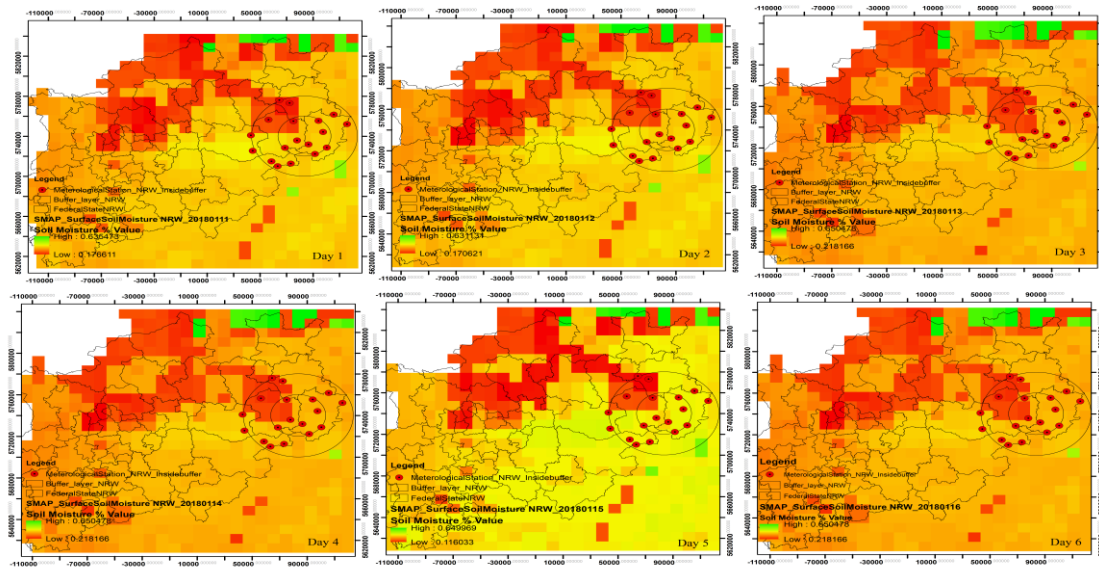


Figure 10: Surface Soil Moisture NRW from 2018-0110 till 2018-0116 (Whole Region)

Annex K : Surface Soil Moisture on Hessen 6 Day series

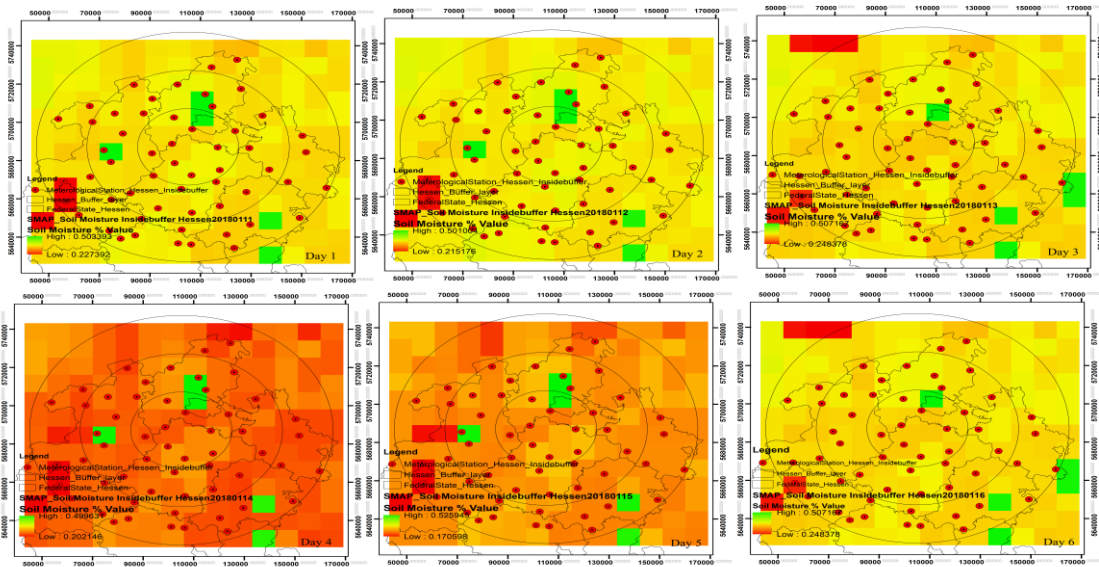


Figure 11: Surface Soil Moisture NRW from 2018-0110 till 2018-0116 (Windthrow Region)

Annex L : Surface Soil Moisture on Hessen 6 Day series

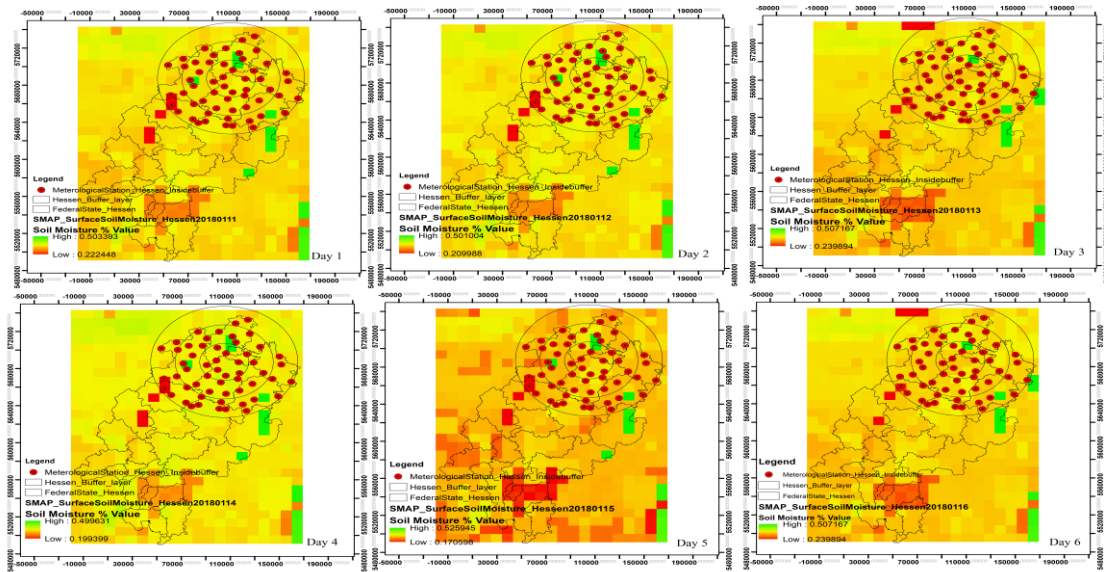


Figure 12: Surface Soil Moisture Hessen from 2018-0110 till 2018-0116 (Whole Region)

Annex M : Rootzone Soil Moisture Brandenburg 6 days series

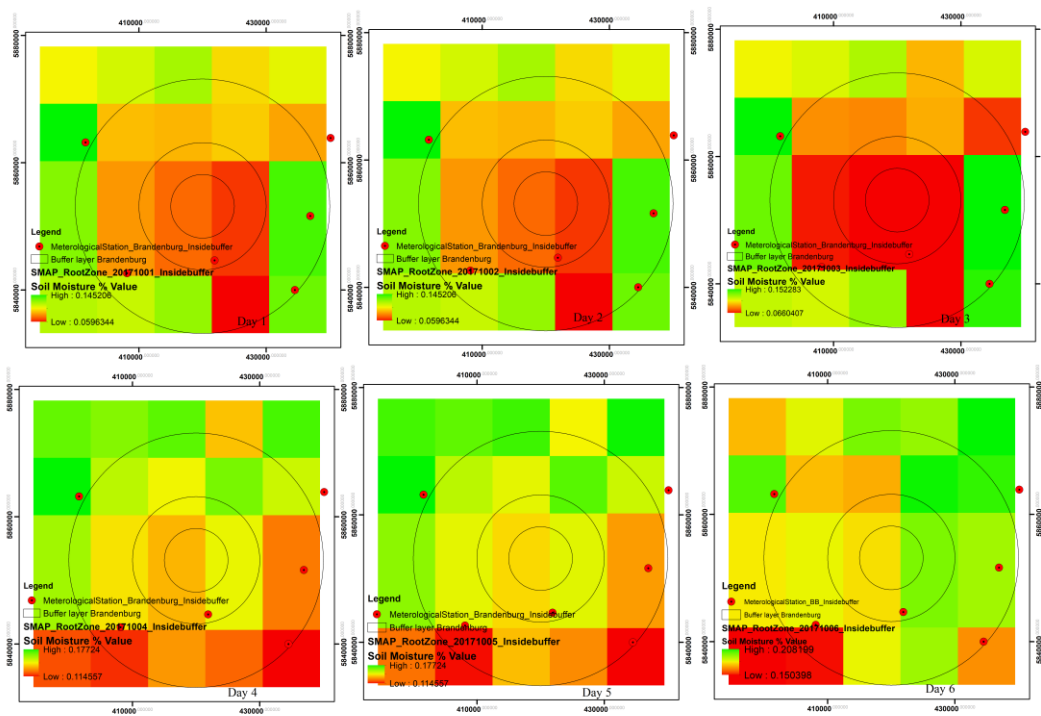


Figure 13: Rootzone Soil moisture Brandenburg from 2017-1001 till 2017-1006 (Windthrow Region)

Annex N: Rootzone Soil Moisture NRW 6 days series

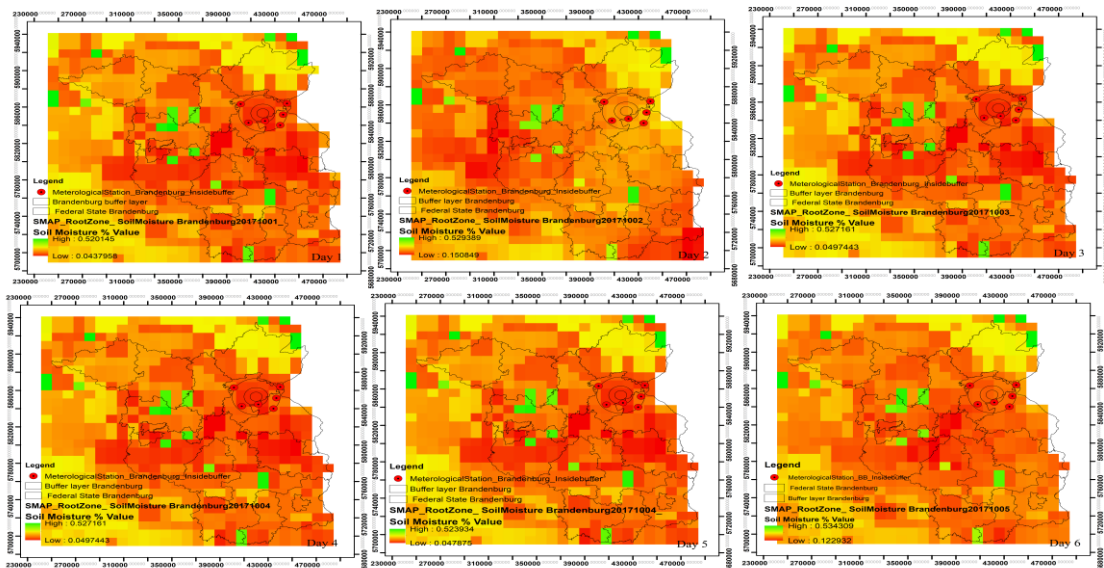


Figure 14: Rootzone Soil Moisture Brandenburg from 2017-1001 till 2017-1006 (Whole Region)

Annex O: Rootzone Soil Moisture NRW 6 days series

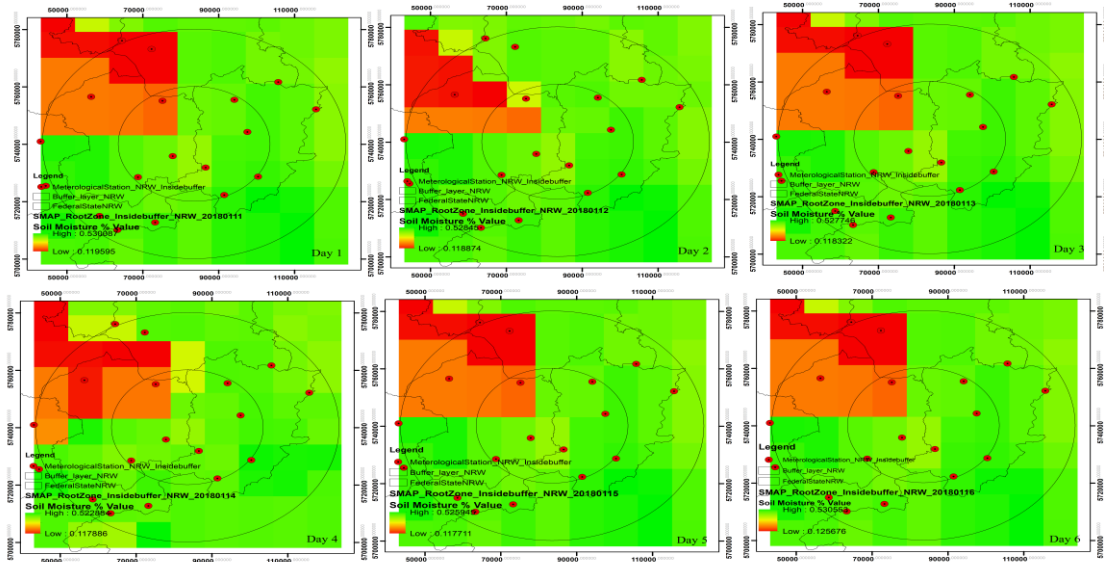


Figure 15: Rootzone Soil Moisture NRW from 2018-0110 till 2018-0116 (Windthrow Region)

Annex P: Rootzone Soil Moisture NRW 6 days series

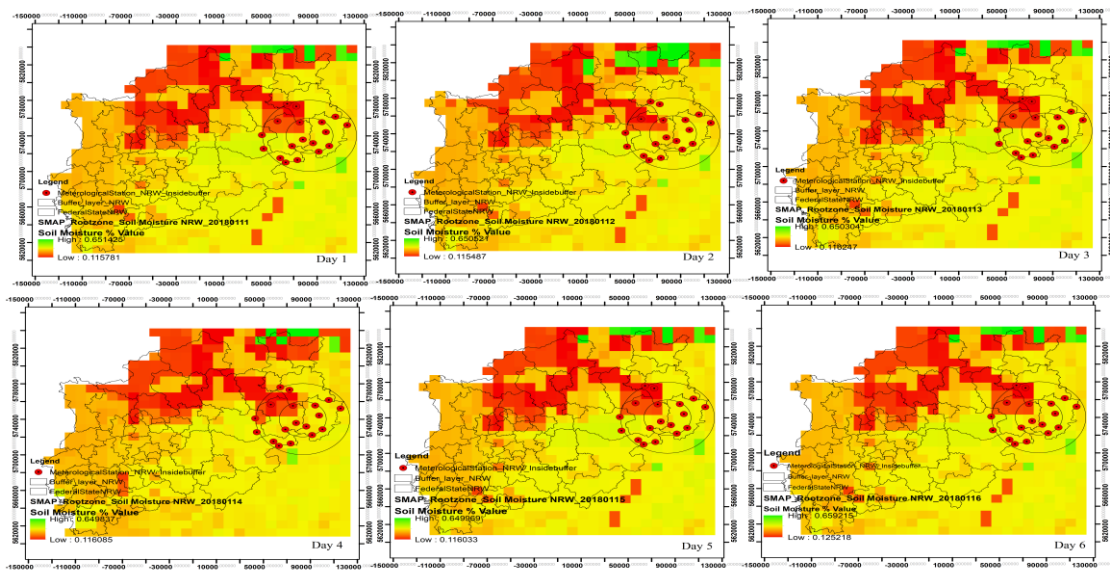


Figure 16: Rootzone Soil Moisture NRW from 2018-0110 till 2018-0116 (Whole Region)

Annex Q: Rootzone Soil Moisture Hessen 6 days series

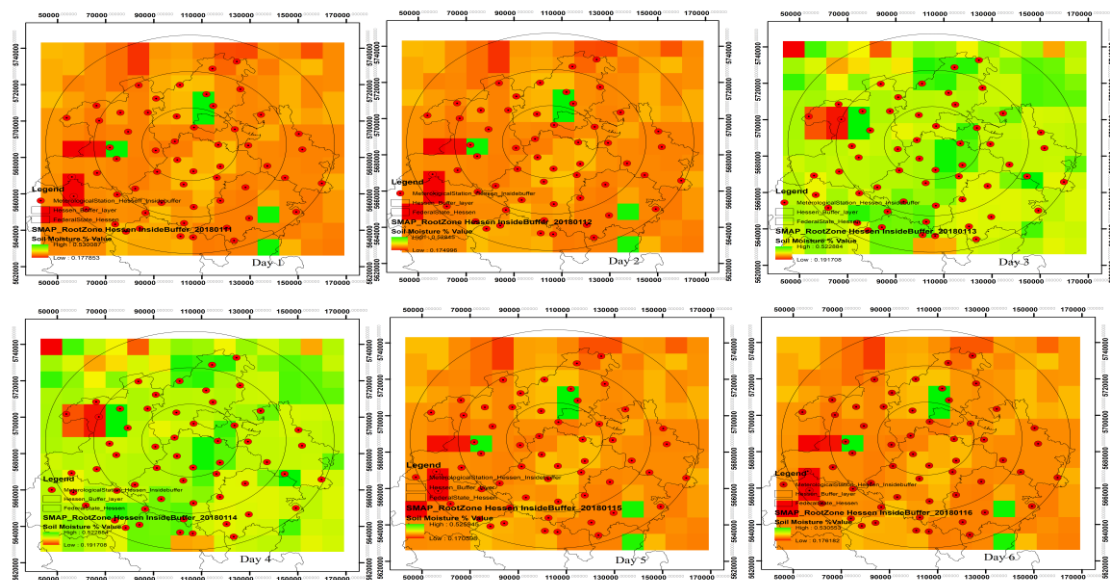


Figure 17: Rootzone Soil Moisture Hessen from 2018-0110 till 2018-0116 (Windthrow Region)

Annex R : Rootzone Soil Moisture Hessen 6 days series

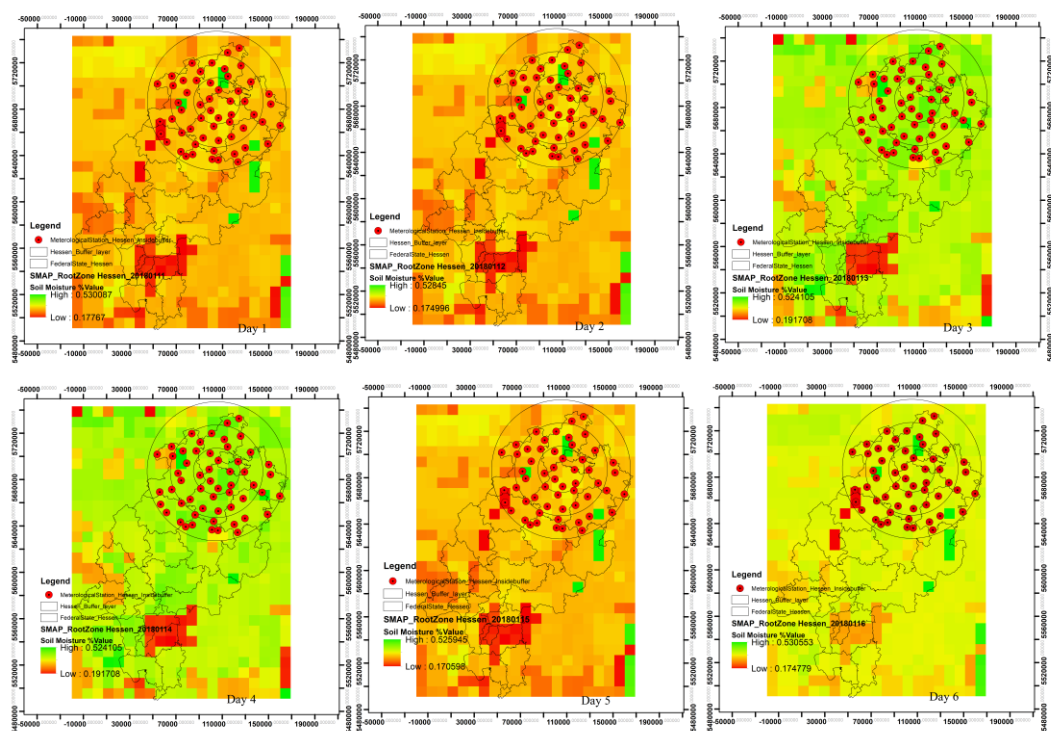


Figure 18: Rootzone Soil Moisture Hessen from 2018-0110 till 2018-0116 (Whole Region)